

Exclusive Ownership of Fiat-Shamir Signatures: ML-DSA, SQIsign, LESS, and More

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Abstract. Exclusive ownership (EO) security is a feature of signature schemes that prevents adversaries from “stealing” an honestly generated signature by finding a new public key which verifies said signature. It is one of the beyond unforgeability features (BUFF) which were declared to be desirable features by NIST. The BUFF transform allows to generically achieve exclusive ownership (and other properties) at the cost of an increased signature size. In this work, we study the EO security of (different variants of) Fiat-Shamir signatures. As our main result, we show that the commonly used variant of Fiat-Shamir signatures (where signatures consist of challenge-response tuples) with λ -bit challenges, can achieve about λ -bit EO security through its implicit usage of the BUFF transform—this presents a significant improvement to existing results that only provide $\lambda/2$ -bit of EO security. This benefit of our result comes without an increase in signature size. For other variants of Fiat-Shamir signatures, we show worse bounds, which nevertheless improve upon existing results. Finally, we apply our results to several signature schemes: SQISign and LESS (both round-2 NIST candidates); ML-DSA (NIST standard); CSI-FISH; and Schnorr signatures. This shows that all these schemes achieve significantly better bounds regarding their EO security compared to existing results.

1 Introduction

Signature schemes allow a user Alice—holding a secret key $\mathbf{sk}$ —to authenticate a message $\mathbf{msg}$ by creating a signature $\mathbf{sig}$. Anyone else—holding Alice’s public key $\mathbf{pk}$ —can then verify that $\mathbf{sig}$ is a valid signature for the message $\mathbf{msg}$. One widely spread method to construct signature schemes is to start with a canonical identification scheme [AABN02]. These are three round protocols, which allow a prover, holding a secret key $\mathbf{sk}$, to convince a verifier, holding the corresponding

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public key $\mathbf{pk}$, of having knowledge of the secret key. The prover sends a commitment $\mathbf{cmt}$, receives a random challenge $\mathbf{ch}$ by the verifier, and then sends a response $\mathbf{rsp}$. The triple $(\mathbf{cmt}, \mathbf{ch}, \mathbf{rsp})$ is called a *transcript*. The Fiat-Shamir transform [FS87] turns such an identification scheme into a signature scheme: the prover—now called the signer—simulates an honest execution of the interactive protocol by locally generating the challenge $\mathbf{ch}$ as $H(\mathbf{cmt}, \mathbf{msg})$,³ where $\mathbf{cmt}$ is the commitment, $\mathbf{msg}$ is the message to be signed, and H is a random oracle. There are, however, different variants,⁴ depending on which parts of the transcript form the signature $\mathbf{sig}$:

- commitment variant ($\mathbf{FS}_{\text{ct}}$): $\mathbf{sig} = (\mathbf{cmt}, \mathbf{rsp})$,
- transcript variant ($\mathbf{FS}_{\text{tr}}$): $\mathbf{sig} = (\mathbf{cmt}, \mathbf{ch}, \mathbf{rsp})$, and
- challenge variant ($\mathbf{FS}_{\text{ch}}$): $\mathbf{sig} = (\mathbf{ch}, \mathbf{rsp})$.

While there have been many works on Fiat-Shamir signatures [OO98, AABN02, PS00, FS87, GQ88, Oka93, Sch91], they typically do not explicitly state which of the variants they are using, simply referring to it as Fiat-Shamir. The different variants are explicitly compared and analyzed in [BSS18].

Historically, signature schemes were required to be unforgeable: it should be infeasible for an adversary Eve to create a valid message-signature pair for Alice’s public key, even after seeing many valid pairs for that key. Fiat-Shamir signatures inherit their unforgeability from the security of the underlying identification scheme. When using signature schemes in more complex protocols, however, one might require more security guarantees from a signature scheme to rule out attacks against the protocol. As of now, these guarantees are known as the *beyond unforgeability features* [CDF⁺21], which we will henceforth refer to as BUFF security notions. They encompass three properties:

1. exclusive ownership (EO),
2. message-bound signatures (MBS), and
3. non-resignability (NR).

The first, exclusive ownership,⁵ ensures that Alice “owns” signatures that she created: Eve cannot find a public key $\mathbf{pk}_E$ —different from Alice’s—that verifies a signature created by Alice. The second, message-bound signatures, ensures that Eve is unable to find a public key $\mathbf{pk}_E$ and a signature $\mathbf{sig}_E$ that verifies two messages $\mathbf{msg}_0$ and $\mathbf{msg}_1$ of Eve’s choice. Such a public key would bypass the non-repudiability: When being accused to have signed message $\mathbf{msg}_i$, Eve can claim to have signed $\mathbf{msg}_{1-i}$ instead. The third, non-resignability, ensures that Eve cannot re-sign an unknown⁶ message after seeing a signature of it.

³Looking ahead, we will actually consider the variant including so-called key prefixing, where the public key $\mathbf{pk}$ is also provided as input to H .

⁴Note that these variants can be derandomized by deriving the random coins for the signing algorithm from a pseudorandom function.

⁵Note that there are different flavors of this property (M-S-UEO, S-UEO, S-CEO, and S-DEO); above we describe the gist underlying all of those.

⁶Formalizing what an unknown message is, turned out to be non-trivial as the original definition of non-resignability was shown to be unachievable [DFHS24].

Achieving BUFF security is surprisingly simple. Rather than just signing the message `msg`, one first hashes the message together with the public key. The resulting hash value is then signed and additionally appended to the signature. This modification is what is now known as the BUFF transform as given in [CDF⁺21]. What makes the BUFF transform so appealing is the fact that it is generic which allows to apply it to virtually any signature scheme. The downside is that it results in larger signatures—due to the appended hash value—which, for schemes with very short signatures, say, SQUISH [AAA⁺24], is undesirable. This disadvantage is further worsened by the fact that the BUFF transform only guarantees $\lambda/2$ -bit of exclusive ownership security for λ -bit hash output length via a collision resistance argument. To achieve λ -bit exclusive ownership security, one needs to append a 2λ -bit hash value to the signature when reasoning via the BUFF transform. On the other hand, Fiat-Shamir signatures—at least the transcript/challenge variant with key prefixing—somewhat implicitly apply the BUFF transform as they include such a hash in the signature—though this hash also includes the commitment. While mentioning this implicit usage of the BUFF transforms, Cremers et al. [CDF⁺21] did not analyze its implications on the EO bounds beyond the bounds given by the generic BUFF transform. This raises the following question:

Can we achieve better bounds for BUFF security of Fiat-Shamir signatures?

1.1 Our Contribution

Firstly, we show a negative result regarding M-S-UEO (malicious-strong universal exclusive ownership) security—the strongest exclusive ownership notion. By giving a birthday attack against M-S-UEO that applies to all Fiat-Shamir variants, we show that one cannot achieve more than $\lambda/2$ bits of M-S-UEO security. This attack relies on so-called weak keys, which verify every signature.⁷

After this, we restrict our focus to the notion S-UEO, which—while being weaker than M-S-UEO—is more important from a practical point of view. Both notions require the adversary to find a public key $\bar{\mathbf{pk}}$ that verifies a signature `sig` valid under a different public key $\mathbf{pk}$. The difference is how the $\mathbf{pk}$ and `sig` are generated: in case of S-UEO (strong universal exclusive ownership), $\mathbf{pk}$ is an honestly generated public key and the adversary obtains `sig` from a signing oracle with the corresponding secret key; in case of M-S-UEO, the adversary can generate $\mathbf{pk}$ and `sig` at will with the only restriction that `sig` has to verify under $\mathbf{pk}$. Thus, from a practical point of view, S-UEO is more relevant as the signing oracle captures the idea that the adversary tries to steal ownership of a signature generated by an honest user.

As our main contribution, we investigate the S-UEO security of Fiat-Shamir signatures. This allows us to answer the aforementioned research question positively, as we can achieve better bounds for some variants of Fiat-Shamir signatures. Our results regarding the different variants are described below.

⁷We leave it as an open problem to provide a bound in the absence of such keys.

$\mathbf{FS}_{\text{ct}}$ (*Commitment Variant*). We show, unsurprisingly, that $\mathbf{FS}_{\text{ct}}$ does not achieve S-UEO security. The main reason is that this variant does not include the challenge ch (which corresponds to the hash value used in the BUFF transform) in the signature. Thus, this variant actually can be viewed as implicitly using one of the Pornin-Stern transforms [PS05], which Cremers et al. [CDF⁺21]—due to the existence of weak keys—showed to be insufficient to achieve S-UEO security.

$\mathbf{FS}_{\text{tr}}$ (*Transcript Variant*). We show that S-UEO security for $\mathbf{FS}_{\text{tr}}$ is upper bounded by $\frac{q_S^2}{|\mathcal{S}_{\text{ct}}|} + \frac{q_H}{2^{|\text{ch}|}}$, for q_S , q_H , and $\mathcal{S}_{\text{ct}}$ being the number of signing queries, random oracle queries, and commitment space, respectively. The first summand is not the dominant one as $|\mathcal{S}_{\text{ct}}| \gg 2^{|\text{ch}|}$ and (online) signing queries are much more costly than (offline) random oracle queries; the ongoing NIST standardization process for post-quantum signature schemes states that security has to hold for adversaries obtaining at most 2^{64} signatures. Therefore, the dominant term is $\frac{q_H}{2^{|\text{ch}|}}$ which improves significantly compared to the bound for the BUFF transform which is $\frac{q_H^2}{2^{|\text{ch}|}}$. Since $\mathbf{FS}_{\text{tr}}$ differs from $\mathbf{FS}_{\text{ct}}$ by additionally appending the hash value to the signature, the difference between these variants is the same as the difference between the BUFF transform and the Pornin-Stern transform.

$\mathbf{FS}_{\text{ch}}$ (*Challenge Variant*). Since the challenge variant $\mathbf{FS}_{\text{ch}}$ does not include the commitment in the signature, it only works for identification schemes that allow to recover the commitment cmt from the challenge-response pair (ch, rsp) . The overall idea for the proof is similar, however, the absence of the commitment in the signature complicates the argument. Since the commitments are recovered from the signature and the public key provided by the adversary, it can happen that the adversary finds a public key for which several, say, t , signatures result in the same recovered commitment. In this case, a random oracle query can have more than a single target. To deal with that, we introduce a new security notion $t\text{-Cmt-CR}_r$ which asks the adversary to find a public key that recovers the same commitment for t honestly generated signatures. Overall, this allows to bound the S-UEO security by $\mathbf{Adv}^{t\text{-Cmt-CR}_r}(\mathcal{B}) + \frac{(t-1)q_H}{2^{|\text{ch}|}}$. In case the adversary cannot find such collisions for the commitments, the dominant term boils down to $\frac{q_H}{2^{|\text{ch}|}}$, as is the case for $\mathbf{FS}_{\text{tr}}$.

$\mathbf{dFS}_{\text{ch}}$ (*Derandomized Challenge Variant*). S-UEO can be further decomposed into S-CEO (strong conservative exclusive ownership) and S-DEO (strong destructive exclusive ownership) security: the difference is whether the adversary has to find a public key that verifies the same message-signature pair as provided by the sign oracle (S-CEO) or find a public key and a *different* message that verifies for one of the signatures provided by the sign oracle (S-DEO). When considering the derandomized version of $\mathbf{FS}_{\text{ch}}$, we give a better bound for S-CEO security. In this case, the problem of colliding commitments vanishes as the adversary can only obtain a single signature per message as signing is derandomized. Based on this, S-CEO security is upper bound by $\frac{q_H}{2^{|\text{ch}|}}$.

Application to Real-World Signature Schemes. We then analyze several signature schemes which apply FS_{ch} . Using our general results above, it remains to argue about the commitment-collision resistance of the schemes. For Schnorr signatures [Sch91], we show that it is always possible to find a 2-commitment collision, while any larger collisions becomes significantly harder. We show that the NIST standard ML-DSA [NIST24a] achieves 2-commitment-collision resistance. For the NIST round-2 candidate SQISIGN [AAA⁺24], we also show the hardness of finding 2-commitment collisions, though based on some heuristic assumptions. Further, we prove that it is hard to find 2-commitment collisions for CSI-FiSh [BKV19]—both in the base variant and its variant with increased challenge space. For the latter case, we require additional properties which are satisfied by some but not all parameter sets. Finally, we show that LESS’ modification to FS_{ch} allows for a simpler proof of S-UEO security due to the usage of random salts.

1.2 Related Work

The concept of signature schemes was introduced in the seminal work by Diffie and Hellman [DH76]. *Full Domain Hash* (FDH) signatures are a class of signatures, which encompass, for instance, the first signature scheme due to Rivest et al. [RSA78]. FDH signatures also exist from quantum-hard assumptions like lattices [GPV08, PFH⁺20]. Unlike other asymmetric cryptographic primitives like public-key encryption, signature schemes can also be realized solely from one-way functions [Lam79, NY89, Rom90]. This type of signatures also appears in quantum-resistant signature schemes [HBG⁺18, NIST24b, BCD⁺23, BCC⁺23]. Another large class of signature schemes involve those constructed via the Fiat-Shamir transform, initially proposed in [FS87] (in the challenge variant) and first proven secure in [PS00] (in the transcript variant). An explicit analysis considering the different variants is given in [BBSS18]. Many proposed signature schemes are constructed via this transform [AABN02, GQ88, Oka93, OO98, Sch91]. Fiat-Shamir signatures also exist from quantum-hard assumptions, e.g., isogenies [CSD⁺23, BKV19] and lattices [NIST24a]—in case of lattices, relying on the so-called *Fiat-Shamir with aborts* framework [Lyu09].

Unforgeability is nowadays the standard security notion signature schemes have to satisfy and was formalized in [GMR88]. Exclusive ownership originates in [PS05] and can be traced back further to [BWM99, MS04]; non-resignability first appeared—though without a formal definition—in [JCCS19]. More recently, Cremers et al. [CDF⁺21] formalized the BUFF notions—incorporating exclusive ownership, non-resignability, and message-bound signatures—and provided a generic transformation which achieves these notions. Since their formalization, the BUFF notions have been considered in various works [ADM⁺24, DFF24, DFH⁺24, DFHS24, DS24, Emu24, KX24].

2 Preliminaries

This section provides the necessary background for this work.⁸ After starting with some notation, we first define canonical identification schemes and signature schemes, followed by a description of Fiat-Shamir signatures. We then provide the relevant BUFF security notions and state a result regarding birthday attacks that we need for our attacks.

Notation. Throughout this work, we write $[k]$ (for an integer k) for the set $\{1, \dots, k\}$. The security parameter is denoted by λ and will be considered an explicit input (in unary) to all algorithms. Adversaries are understood as efficient, i.e., probabilistic polynomial-time, algorithms wrt λ .

Identification Schemes. Below we define canonical identification schemes, following the description of [AABN02].

Definition 1. A (canonical) identification scheme Σ consists of four efficient algorithms $(\text{KeyGen}, \text{Cmt.G}, \text{Rsp.G}, \text{Rsp.V})$ where:

$\text{KeyGen}(1^\lambda) \rightarrow (\text{sk}, \text{pk})$ is the key generation algorithm. It takes as input a security parameter and outputs a key pair (sk, pk) . We make use of the common convention that the secret key contains the public key as well.

$\text{Cmt.G}(\text{sk}) \rightarrow \text{cmt}$ is the commitment generation algorithm. It takes as input a secret key sk ⁹ and outputs a commitment cmt .

$\text{Rsp.G}(\text{sk}, \text{cmt}, \text{ch}) \rightarrow \text{rsp}$ is the response generation algorithm. It takes as input a secret key sk , a commitment cmt , and a challenge ch , and it outputs a response rsp .

$\text{Rsp.V}(\text{pk}, \text{cmt}, \text{ch}, \text{rsp}) \rightarrow v$ is the response verification algorithm. It takes as input a secret key sk , a commitment cmt , a challenge ch , and a response rsp , and outputs a bit v .

A triple $(\text{cmt}, \text{ch}, \text{rsp})$ is called a transcript. An identification scheme is said to be correct, if the response verification algorithm accepts, i.e., outputs 1, on honestly generated transcripts (with overwhelming probability). We let $\mathcal{S}_{\text{ct}}$ and $\mathcal{S}_{\text{ch}}$ denote the commitment space and challenge space, respectively. Throughout this work, the challenge space will typically contain bit strings of a certain length, i.e., $|\mathcal{S}_{\text{ch}}| = 2^{|\text{ch}|}$. Further note that typically $|\mathcal{S}_{\text{ch}}| \leq |\mathcal{S}_{\text{ct}}|$ holds.

For several identification schemes, there is a fifth algorithm Cmt.R where:

$\text{Cmt.R}(\text{pk}, \text{ch}, \text{rsp}) \rightarrow \text{cmt}$ is the commitment recovery algorithm. It takes as input a public key pk , a challenge ch , and a response rsp , and outputs a commitment cmt .

⁸Specific background for the signature schemes analyzed in Section 4 is provided in the respective sections.

⁹Note that commitment generation typically only depends on the public key contained in the secret key, e.g., for ML-DSA, or is independent of the key, e.g., for Schnorr signatures.

For schemes that have a commitment recovery algorithm, we write Σ as a five-tuple $(\text{KeyGen}, \text{Cmt.G}, \text{Rsp.G}, \text{Rsp.V}, \text{Cmt.R})$.

Signature Schemes. Below we define signature schemes.

Definition 2. A signature scheme S is a collection of three efficient algorithms $(\text{KeyGen}, \text{Sign}, \text{Verify})$ where:

$\text{KeyGen}(1^\lambda) \rightarrow (\text{sk}, \text{pk})$ is the key generation algorithm. It takes as input a security parameter and outputs a key pair (sk, pk) .

$\text{Sign}(\text{sk}, \text{msg}) \rightarrow \text{sig}$ is the signing algorithm. It takes as input a secret key sk and a message msg , and outputs a signature sig .

$\text{Verify}(\text{pk}, \text{msg}, \text{sig}) \rightarrow \text{v}$ is the verification algorithm. It takes as input a public key pk , a message msg , and a signature sig , and it outputs a bit v .

A signature scheme is said to be correct, if, for any $(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$ and any message msg , $\text{Verify}(\text{pk}, \text{msg}, \text{Sign}(\text{sk}, \text{msg})) = 1$ holds with overwhelming probability.

Fiat-Shamir Signatures. The Fiat-Shamir paradigm [FS87] allows to transform an identification scheme into a signature scheme. The key idea is to replace the generation of a random challenge—done by the verifier in an interactive fashion in an identification scheme—by a random oracle H . This enables the signer to non-interactively generate a random challenge (exploiting the fact that the signer has no control over the output of the random oracle), which can later be verified by the verifier. Over the years, three different variants of the Fiat-Shamir paradigm have emerged:

- FS_{ct} (commitment variant): In this variant, the signature consists of the commitment and the response.
- FS_{tr} (transcript variant): In this variant, the signature consists of the whole transcript, i.e., commitment, challenge, and response.
- FS_{ch} (challenge variant): In this variant, the signature consists of the challenge and the response.

The former two (FS_{ct} and FS_{tr}) work for any identification scheme, whereas the latter (FS_{ch}) only works for identification schemes that come with a commitment recovery algorithm Cmt.R .

The different variants are considered in various works, though they are typically simply called Fiat-Shamir signatures without specifying the exact variant: FS_{tr} has been investigated by Pointcheval and Stern [PS00]; FS_{ct} has been studied by Ohta and Okamoto [OO98] as well as Abdalla et al. [AABN02]; and FS_{ch} is considered by Guillou and Quisquater [GQ88], Okamoto [Oka93], Schnorr [Sch91], as well as in the original work by Fiat and Shamir [FS87]. A thorough analysis including all variants was done by Backendal et al. [BBSS18].

Fig. 1 displays the pseudocode for FS_{ct} , FS_{tr} , and FS_{ch} . Note that we consider variants, which include so-called *key prefixing* [Ber15], meaning that the

$\text{FS}_{\text{ct}}.\text{Sign}(\text{sk}, \text{msg})$	$\text{FS}_{\text{tr}}.\text{Sign}(\text{sk}, \text{msg})$	$\text{FS}_{\text{ch}}.\text{Sign}(\text{sk}, \text{msg})$
$\text{cmt} \leftarrow \text{\$ Cmt.G}(\text{sk})$	$\text{cmt} \leftarrow \text{\$ Cmt.G}(\text{sk})$	$\text{cmt} \leftarrow \text{\$ Cmt.G}(\text{sk})$
$\text{ch} \leftarrow \text{H}(\text{pk}, \text{cmt}, \text{msg})$	$\text{ch} \leftarrow \text{H}(\text{pk}, \text{cmt}, \text{msg})$	$\text{ch} \leftarrow \text{H}(\text{pk}, \text{cmt}, \text{msg})$
$\text{rsp} \leftarrow \text{Rsp.G}(\text{sk}, \text{cmt}, \text{ch})$	$\text{rsp} \leftarrow \text{Rsp.G}(\text{sk}, \text{cmt}, \text{ch})$	$\text{rsp} \leftarrow \text{Rsp.G}(\text{sk}, \text{cmt}, \text{ch})$
$\text{sig} \leftarrow (\text{cmt}, \text{rsp})$	$\text{sig} \leftarrow (\text{cmt}, \text{ch}, \text{rsp})$	$\text{sig} \leftarrow (\text{ch}, \text{rsp})$
return sig	return sig	return sig
$\text{FS}_{\text{ct}}.\text{Verify}(\text{pk}, \text{msg}, \text{sig})$	$\text{FS}_{\text{tr}}.\text{Verify}(\text{pk}, \text{msg}, \text{sig})$	$\text{FS}_{\text{ch}}.\text{Verify}(\text{pk}, \text{msg}, \text{sig})$
$(\text{cmt}, \text{rsp}) \leftarrow \text{sig}$	$(\text{cmt}, \text{ch}, \text{rsp}) \leftarrow \text{sig}$	$(\text{ch}, \text{rsp}) \leftarrow \text{sig}$
$\text{ch} \leftarrow \text{H}(\text{pk}, \text{cmt}, \text{msg})$	if $\text{ch} \neq \text{H}(\text{pk}, \text{cmt}, \text{msg})$	$\text{cmt} \leftarrow \text{Cmt.R}(\text{pk}, \text{ch}, \text{rsp})$
$\text{v} \leftarrow \text{Rsp.V}(\text{pk}, \text{cmt}, \text{ch}, \text{rsp})$	return 0	if $\text{cmt} = \perp$
return v	$\text{v} \leftarrow \text{Rsp.V}(\text{pk}, \text{cmt}, \text{ch}, \text{rsp})$	return 0
	return v	if $\text{ch} \neq \text{H}(\text{pk}, \text{cmt}, \text{msg})$
		return 0
		return 1

Fig. 1: **Left:** Fiat-Shamir commitment variant (FS_{ct}). **Middle:** Fiat-Shamir transcript variant (FS_{tr}). **Right:** Fiat-Shamir challenge variant (FS_{ch}).

challenge computation does include the public key. While initially developed as a method to protect against multi-user attacks, it also turned out to be crucial for BUFF security.

Beyond Unforgeability Features. In this work, we focus on exclusive ownership, which encompasses strong universal exclusive ownership (S-UEO) and the generally stronger version called malicious-strong universal exclusive ownership (M-S-UEO)—the former can be further distinguished into strong conservative exclusive ownership (S-CEO) and strong destructive exclusive ownership (S-DEO). Together with the notions of message-bound signatures (MBS) and non-resignability (NR), these notions form the so-called beyond unforgeability features. Fig. 2 shows game S-UEO: an adversary $\mathcal{A}$ gets a public key pk and access to a signing oracle with the corresponding secret key. To win the game, $\mathcal{A}$ has to output a new public key $\overline{\text{pk}}$ (different from pk) and a message-signature pair $(\overline{\text{msg}}, \overline{\text{sig}})$ such that the pair verifies under $\overline{\text{pk}}$ and $\overline{\text{sig}}$ was the response of one of the signing queries by $\mathcal{A}$. This game unifies the two underlying notions S-CEO and S-DEO. To obtain these two games, one only needs to adapt the final check for the output of the adversary. For S-CEO, not just $\overline{\text{sig}}$ has to be the result of a signing query but the pair $(\overline{\text{msg}}, \overline{\text{sig}})$ has to be from one signing query. For S-DEO, $\overline{\text{sig}}$ has to be a signature obtained from a signing query on a message msg different from $\overline{\text{msg}}$. Below we provide the formal definition

Definition 3. Let $\text{S} = (\text{KeyGen}, \text{Sign}, \text{Verify})$ be a signature scheme and the game $\text{X} \in \{\text{S-UEO}, \text{S-CEO}, \text{S-DEO}\}$ be defined as in Fig. 2. For any adversary

Game S-UEO	Game M-S-UEO
$Q \leftarrow \emptyset$	$(pk, \overline{pk}, msg, \overline{msg}, sig) \leftarrow \mathcal{A}()$
$(sk, pk) \leftarrow \Sigma.KeyGen()$	$v_1 \leftarrow S.Verify(pk, msg, sig)$
$(\overline{pk}, \overline{msg}, \overline{sig}) \leftarrow \mathcal{A}^{S.Sign(sk, \cdot)}(\overline{pk})$	$v_2 \leftarrow S.Verify(\overline{pk}, \overline{msg}, \overline{sig})$
$v_1 \leftarrow S.Verify(pk, \overline{msg}, \overline{sig})$	return $\llbracket v_1 = 1 \wedge v_2 = 1 \wedge \overline{pk} \neq pk \rrbracket$
$v_2 \leftarrow Valid(\overline{msg}, \overline{sig})$	$Valid(\overline{msg}, \overline{sig}) \quad // \text{ S-UEO}$
return $\llbracket v_1 = 1 \wedge v_2 = 1 \wedge \overline{pk} \neq pk \rrbracket$	
Oracle $Sign(sk, msg) \quad // \text{ S-UEO}$	if $(\overline{msg}, \overline{sig}) \in Q$
$sig \leftarrow S.Sign(sk, msg)$	return 1 $// \text{ S-CEO}$
$Q \leftarrow Q \cup \{msg, sig\}$	if $\exists msg \neq \overline{msg} \text{ s.t. } (msg, \overline{sig}) \in Q$
return sig	return 1 $// \text{ S-DEO}$
	return 0

Fig. 2: Exclusive ownership security games S-UEO and M-S-UEO for a signature scheme $S = (KeyGen, Sign, Verify)$. For ease of exposition, we give the validity check ($Valid$) separately for S-CEO/S-DEO security. The unified check for S-UEO is equivalent to checking whether $\overline{sig} \in Q$.

$\mathcal{A}$, making $q_S \geq 1$ signing queries to $Sign$ and q_H random oracle queries, its $X \in \{S\text{-UEO}, S\text{-CEO}, S\text{-DEO}\}$ advantage is defined as

$$\mathbf{Adv}_S^X(\mathcal{A}) := \Pr[X(\mathcal{A}) \rightarrow 1].$$

Regarding the number of signing queries, we follow the call for digital signature proposals by NIST [NIST22] and restrict our focus to $q_S \leq 2^{64}$.

The stronger form of exclusive ownership is given by game M-S-UEO as displayed in Fig. 2. It works similar to S-UEO except that the adversary is not restricted to signatures from a signing oracle using an honestly generated secret key. Instead, the adversary is challenged to output a signature sig alongside with two tuples (msg, pk) and $(\overline{msg}, \overline{pk})$ of message and public key such that the signature verifies the messages under the respective public keys. While the public keys have to be different, i.e., $pk \neq \overline{pk}$, there is no requirements on the message, in particular, $\mathcal{A}$ can set $msg = \overline{msg}$. Below we provide the formal definition.

Definition 4. Let $S = (KeyGen, Sign, Verify)$ be a signature scheme and the game M-S-UEO be defined as in Fig. 2. For any adversary $\mathcal{A}$, making q_H random oracle queries, its M-S-UEO advantage is defined as

$$\mathbf{Adv}_S^{M\text{-S-UEO}}(\mathcal{A}) := \Pr[M\text{-S-UEO}(\mathcal{A}) \rightarrow 1].$$

Generalized Birthday Attack. One of our attacks relies on the generalized birthday attack due to Wagner [Wag02], which we recall here. The lemma below is taken verbatim from [KSW24].

Lemma 5. *Consider two lists L_1, L_2 of elements drawn uniformly and independently at random from $\{0, 1\}^\tau$. We denote the size of L_1 and L_2 by l_1 and l_2 , respectively, and define $l = l_1 + l_2$. Then one finds $x_k \in L_1$ and $x_j \in L_2$ such that $x_k \oplus x_j = 0$ with a probability of at least*

$$\left(1 - \exp\left(\frac{-l(l-1)}{2^{\tau+1}}\right)\right) \cdot \frac{2l_1l_2}{l^2 - l}.$$

3 Exclusive Ownership of Fiat-Shamir Signatures

In this section, we analyze the exclusive ownership properties of the different Fiat-Shamir variants. In [Section 3.1](#), we start with our negative result regarding M-S-UEO security of Fiat-Shamir signatures, affecting all variants, and discuss the practical relevance of S-UEO security. [Section 3.2](#) and [Section 3.3](#) cover the commitment variant and the transcript variant, respectively. The challenge variant is considered in [Section 3.4](#) while [Section 3.5](#) is concerned with its deterministic version.

3.1 Malicious Exclusive Ownership and Weak Keys

We start by introducing an identification scheme with a weak key that we will use several times. Consider an arbitrary identification scheme Σ and transform it as follows: (1) fix a weak public key pk_w ; (2) modify the verification algorithm to output 1 whenever the input public key is pk_w ; (3) in case the identification scheme has a commitment recovery algorithm, fix a weak commitment cmt_w ¹⁰ and modify the algorithm to always output cmt_w on input public key pk_w . An illustration is given in [Fig. 3](#). Note that this identification is sufficient to achieve standard unforgeability; the reason is that the weak key occurs only with negligible probability for honestly generated keys in unforgeability security notions.

The following theorem describes an attack (using about $2^{\frac{|\text{ch}|}{2}}$ queries to the random oracle) against the M-S-UEO security of FS_{tr} using a weak key.

Theorem 6. *Let Σ_w be an identification scheme as described in [Fig. 3](#) and H be a random oracle. Consider the signature scheme $\text{FS}_{\text{tr}}[\Sigma_w, \text{H}]$. Then there is an adversary $\mathcal{A}$, making $q_H = 2^{\frac{|\text{ch}|}{2} + 1} + 2$ queries to H such that*

$$\text{Adv}_{\text{FS}_{\text{tr}}[\Sigma_w]}^{\text{M-S-UEO}}(\mathcal{A}) \geq \frac{3}{8}.$$

Proof. Adversary $\mathcal{A}$ generates an honest key pair (sk, pk) and picks the weak key pk_w . Furthermore, $\mathcal{A}$ picks an arbitrary commitment cmt ,¹¹ and $k = 2^{\frac{|\text{ch}|}{2} + 1}$ messages $\text{msg}_1, \dots, \text{msg}_k$. It then makes $2^{\frac{|\text{ch}|}{2} + 1}$ queries $(\text{pk}, \text{cmt}, \text{msg}_i)$ to H and

¹⁰This is only required when extending the result to FS_{ch} .

¹¹When extending the result to FS_{ch} , $\mathcal{A}$ has to pick the commitment cmt_w .

$\Sigma_w.\text{Resp.V}(\text{pk}, \text{cmt}, \text{ch}, \text{rsp})$	$\Sigma_w.\text{Cmt.R}(\text{pk}, \text{ch}, \text{rsp})$
if $\text{pk} = \text{pk}_w$	if $\text{pk} = \text{pk}_w$
return 1	return cmt_w
$v \leftarrow \Sigma.\text{Resp.V}(\text{pk}, \text{cmt}, \text{ch}, \text{rsp})$	$\text{cmt} \leftarrow \Sigma.\text{Cmt.R}(\text{pk}, \text{ch}, \text{rsp})$
return v	return cmt

Fig. 3: Identification scheme with a weak public key pk_w that accepts any transcript. Furthermore, when considering the challenge variant, the commitment recovery algorithm is modified so that the $\text{Cmt.R}(\text{pk}, \cdot, \cdot)$ outputs a particular commitment cmt_w irrespective of the other inputs. The other algorithms, i.e., KeyGen , Cmt.G , and Resp.G are unaltered.

stores the resulting hash values in a list L_1 . Subsequently, $\mathcal{A}$ makes another $2^{\frac{|\text{ch}|}{2}} + 1$ queries $(\text{pk}_w, \text{cmt}, \text{msg}_i)$, storing the resulting hash values in another list L_2 . By Lemma 5, there is a collision between the elements in list L_1 and the elements in list L_2 with probability at least

$$\left(1 - \exp\left(-\frac{\left(2^{\frac{|\text{ch}|+2}{2}} + 2\right)\left(2^{\frac{|\text{ch}|+2}{2}} + 1\right)}{2^{|\text{ch}|+1}}\right)\right) \cdot \frac{2^{|\text{ch}|+1}}{2^{|\text{ch}|+2} - 2^{\frac{|\text{ch}|}{2}+1}}.$$

It holds that

$$\frac{-\left(2^{\frac{|\text{ch}|+2}{2}} + 2\right)\left(2^{\frac{|\text{ch}|+2}{2}} + 1\right)}{2^{|\text{ch}|+1}} \leq \frac{-\left(2^{\frac{|\text{ch}|+2}{2}}\right)\left(2^{\frac{|\text{ch}|+2}{2}}\right)}{2^{|\text{ch}|+1}} \leq \frac{-2^{|\text{ch}|+2}}{2^{|\text{ch}|+1}} = -2$$

and

$$\frac{2^{|\text{ch}|+1}}{2^{|\text{ch}|+2} - 2^{\frac{|\text{ch}|}{2}+1}} \geq \frac{1}{2}.$$

Thus, we have a collision with probability at least $(1 - \exp(-2)) \cdot \frac{1}{2} \geq \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$. Let i and j denote the indices such that $\text{ch} := \text{H}(\text{pk}, \text{cmt}, \text{msg}_i) = \text{H}(\text{pk}_w, \text{cmt}, \text{msg}_j)$. Finally, adversary $\mathcal{A}$ computes $\text{rsp} \leftarrow \Sigma_w.\text{Resp.G}(\text{sk}, \text{cmt}, \text{ch})$, i.e., it honestly computes the response using the honestly generated secret key, and outputs $(\text{pk}, \text{pk}_w, \text{msg}_i, \text{msg}_j, \text{sig} = (\text{cmt}, \text{ch}, \text{rsp}))$.

Observe that the signature sig verifies under the public key pk as it was generated honestly using the corresponding secret key sk . The signature sig also verifies under the weak public key as (1) the messages were chosen to result in the challenge ch from the signature and (2) the resulting transcript is accepted as pk_w accepts any transcript. $\square$

Extension to FS_{ct} and FS_{ch} . The theorem above shows that FS_{tr} does not guarantee M-S-UEO security beyond birthday attacks on the random oracle. The attack extends to the other variants FS_{ct} and FS_{ch} . When considering FS_{ct} , the

mere difference is that the challenge ch is recomputed from public key, commitment, and message, rather than being part of the signature which has no effect on the argument. Regarding FS_{ch} , the difference is that the commitment is recovered from challenge and response. For the honestly generated public key, correctness ensures that the initially chosen commitment is recovered; the weak public key, on the other hand, always recovers the weak commitment cmt_w , which is why $\mathcal{A}$ has to make its random oracle queries using cmt_w . Observe that the attack is entirely offline: adversary $\mathcal{A}$ searches a collision for the random oracle and—once such a collision is found—generates an honest signature for the message-commitment pair that leads to the collision.

Non-Transferability to S-UEO. One might think that the attack extends to S-UEO by simply replacing the first batch of random oracle queries (those to build the first list) with signing queries. However, there is a catch: In the attack, the adversary makes all random oracle queries on *the same* commitment cmt . In case of S-UEO, these commitments would be chosen by the signing oracle and each query would result in a different commitment. Thus, the strategy from the above theorem cannot be applied for an S-UEO attack.

Open Question. Note that the attack from [Theorem 6](#) relies on a weak key that verifies any transcript. This leaves the possibility that better bounds for M-S-UEO security might be possible when the underlying identification scheme does not exhibit such keys. Looking ahead, the schemes we analyze do not have weak keys, which would make their absence a plausible assumption. We thus leave the open problem of finding a similar attack or proving a better bound in case there are no weak keys.

Practical Relevance. When comparing the two notions M-S-UEO and S-UEO, the gist of both is that $\mathcal{A}$ needs to find a second public key that verifies a signature, which is valid for a different public key. The core difference is how the signature is generated. In case of M-S-UEO, the adversary can generate the signature at will, whereas in case of S-UEO, the adversary is restricted to use honestly generated signatures obtained from a signing oracle. From a practical point of view, this means that S-UEO should be sufficient as $\mathcal{A}$ is interested in claiming ownership of a signature from a different user. As an example, consider the attack proposed against the Let’s Encrypt protocol [[Aye15](#)]: here the adversary tries to claim ownership of a domain by providing Let’s Encrypt with a public key that verifies the signature put on the server by the actual owner. This scenario is already covered by S-UEO using honestly generated signatures. In the remainder of this work, we restrict our focus to S-UEO security.

3.2 Fiat-Shamir (Commitment Variant)

The commitment variant of Fiat-Shamir, FS_{ct} , does not achieve S-UEO security. The problem is due to weak keys and the fact that the challenge is not part of the signature for this variant. The theorem below, which makes use of the weak identification scheme as described in [Fig. 3](#), formalizes this.

Theorem 7. *Let Σ_w be as above and H be a random oracle. Consider the signature scheme $\text{FS}_{\text{ct}}[\Sigma_w, H]$. Then there is an adversary $\mathcal{A}$, making a single query to Sign , such that*

$$\text{Adv}_{\text{FS}_{\text{ct}}[\Sigma_w]}^{\text{S-UEO}}(\mathcal{A}) = 1.$$

Proof. Adversary $\mathcal{A}$ receives a public key pk as input. It picks an arbitrary message msg and queries it to the signing oracle. Let $\text{sig} = (\text{cmt}, \text{rsp})$ be the resulting signature. Adversary $\mathcal{A}$ then outputs $(\text{pk}_w, \text{msg}, \text{sig})$.

To verify validity of the signature, observe that pk_w verifies every transcript. This means that $\text{Rsp.V}(\text{pk}_w, (\text{cmt}, \cdot, \text{rsp}))$ will output 1 regardless of the challenge (output of $H(\text{pk}_w, \text{cmt}, \text{msg})$). Furthermore, the second check in game S-UEO succeeds as the signature sig was returned by the signing oracle. $\square$

By hierarchy, the same attack also extends to M-S-UEO, where the adversary simply generates pk honestly (alongside the corresponding sk) and locally computes the signature sig (using sk). This shows that there are better attacks against the M-S-UEO security of FS_{ct} than the one following from Theorem 6.

Note that FS_{ct} (as described in Fig. 1) implicitly applies one of the Pornin-Stern transforms [PS05] by hashing both the public key and the message to obtain the challenge. However, since the signature does not contain the hash value explicitly, it fails to achieve S-UEO security. We emphasize that this observation is not novel as it already appears in [CDF⁺21]; our example here corresponds to the observation in [CDF⁺21] that the Pornin-Stern transform fails to achieve S-UEO security when weak keys do exist.

3.3 Fiat-Shamir (Transcript Variant)

For the transcript variant of Fiat-Shamir, FS_{tr} , the signature consists of the whole transcript. Compared to the commitment variant FS_{ct} , the signature therefore additionally contains the challenge. One can therefore view FS_{ct} as a signature scheme using the Pornin-Stern transform while FS_{tr} uses the BUFF transform. Therefore, it is not surprising that the transcript variant achieves S-UEO security—at least when arguing via the collision resistance of the used hash function—, which follows from the results in [CDF⁺21]. Interestingly though, we can prove a better bound for S-UEO security: instead of a collision bound $\frac{q_H^2}{|\mathcal{S}_{\text{ch}}|}$, we get a preimage bound $\frac{q_H}{|\mathcal{S}_{\text{ch}}|}$ for the random oracle queries plus a collision bound for the commitments $\frac{q_S^2}{2|\mathcal{S}_{\text{ct}}|}$. Considering that online signing queries are much more expensive and the fact that $|\mathcal{S}_{\text{ct}}| \gg |\mathcal{S}_{\text{ch}}|$, our bound improves significantly compared to [CDF⁺21].

Theorem 8. *Let Σ be an identification protocol and H be a random oracle. Consider the signature scheme $\text{FS}_{\text{tr}}[\Sigma, H]$. Then for any S-UEO adversary $\mathcal{A}$, making q_S queries to the signing oracle and q_H queries to H , we have*

$$\text{Adv}_{\text{FS}_{\text{tr}}[\Sigma, H]}^{\text{S-UEO}}(\mathcal{A}) \leq \frac{q_S^2}{2|\mathcal{S}_{\text{ct}}|} + \frac{q_H}{2^{|\text{ch}|}}.$$

Proof. Let $\mathcal{A}$ be an adversary playing S-UEO. Let further Cmt-Coll denote the event that there are two signing queries by $\mathcal{A}$ that result in signatures which agree in the commitment. A simple calculation yields

$$\begin{aligned} \text{Adv}_S^{\text{S-UEO}}(\mathcal{A}) &= \Pr[\mathcal{A} \text{ wins S-UEO}] \\ &= \Pr[\mathcal{A} \text{ wins S-UEO} \wedge \text{Cmt-Coll}] + \Pr[\mathcal{A} \text{ wins S-UEO} \wedge \neg\text{Cmt-Coll}] \\ &\leq \Pr[\text{Cmt-Coll}] + \Pr[\mathcal{A} \text{ wins S-UEO} \wedge \neg\text{Cmt-Coll}] \\ &\leq \Pr[\text{Cmt-Coll}] + \Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg\text{Cmt-Coll}]. \end{aligned}$$

To bound the first term, i.e., $\Pr[\text{Cmt-Coll}]$, let Cmt-Coll_i denote the event that Cmt-Coll occurs for the i -th signing query by $\mathcal{A}$. Since the signing oracle chooses the commitment uniformly at random from $\mathcal{S}_{\text{ct}}$ for each query, we obtain

$$\Pr[\text{Cmt-Coll}] \leq \sum_{i=1}^{q_S} \Pr[\text{Cmt-Coll}_i] = \sum_{i=0}^{q_S-1} \frac{i}{|\mathcal{S}_{\text{ct}}|} \leq \frac{q_S^2}{2|\mathcal{S}_{\text{ct}}|}.$$

Next, we bound the second term, i.e., $\Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg\text{Cmt-Coll}]$. For this, let $(\text{msg}_1, \text{sig}_1 = (\text{cmt}_1, \text{ch}_1, \text{rsp}_1)), \dots, (\text{msg}_{q_S}, \text{sig}_{q_S} = (\text{cmt}_{q_S}, \text{ch}_{q_S}, \text{rsp}_{q_S}))$ denote the q_S message-signature pairs that $\mathcal{A}$ receives. Conditioning on $\neg\text{Cmt-Coll}$, we know that all signing queries of $\mathcal{A}$ resulted in signatures with distinct commitments, i.e., for all $i \neq j$ we have $\text{cmt}_i \neq \text{cmt}_j$. In order to win game S-UEO, $\mathcal{A}$ has to find a message $\overline{\text{msg}}$ (not necessarily different from the queried messages) and a different public key $\overline{\text{pk}}$, such that any of the received signatures verifies $\overline{\text{msg}}$ under $\overline{\text{pk}}$. For this check to pass, it must hold that $H(\overline{\text{pk}}, \text{cmt}_i, \overline{\text{msg}}) = \text{ch}_i$ for some $i \in \{1, \dots, q_S\}$. This means that each random oracle query by $\mathcal{A}$ has exactly one target value to hit: a query $(\cdot, \text{cmt}_i, \cdot)$ has the target value ch_i . Since each query has success probability of at most $\frac{1}{2^{|\text{ch}|}}$, it follows that

$$\Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg\text{Cmt-Coll}] \leq \frac{q_H}{2^{|\text{ch}|}}.$$

Putting everything together yields

$$\begin{aligned} \text{Adv}_S^{\text{S-UEO}}(\mathcal{A}) &\leq \Pr[\text{Cmt-Coll}] + \Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg\text{Cmt-Coll}] \\ &= \frac{q_S^2}{2|\mathcal{S}_{\text{ct}}|} + \frac{q_H}{2^{|\text{ch}|}}, \end{aligned}$$

which concludes the proof. $\square$

One might argue that the above theorem is irrelevant since every (Fiat-Shamir-based) signature scheme uses either FS_{ct} or FS_{ch} as both result in shorter signatures than FS_{tr} . Given our negative result regarding the S-UEO security of FS_{ct} (cf. [Theorem 7](#)), the above theorem shows that simply including also the challenge to the signature, that is, switching from FS_{ct} to FS_{tr} , is sufficient to achieve S-UEO security. Furthermore, it shows that one achieves about λ -bit of S-UEO compared to $\frac{\lambda}{2}$ -bit when using the existing result for the BUFF transform [\[CDF⁺21\]](#).

3.4 Fiat-Shamir (Challenge Variant)

Out of all Fiat-Shamir variants, the challenge variant, FS_{ch} , is the favored one in practice as it typically results in the smallest signatures. The reason is that the challenge is generally much smaller than the commitment. This variant is used by several schemes, e.g., the lattice-based, NIST-standardized signature scheme ML-DSA [NIST24a]; the isogeny-based signatures SQIsign [CSD⁺23] and CSI-FiSh [BKV19]; the code-based signature scheme LESS [BBB⁺24b]; and the discrete logarithm-based signature scheme by Schnorr [Sch91].

Similar to FS_{tr} , we can show positive results for FS_{ch} since the hash value is part of the signature. Recall that for FS_{ch} , verification works by recovering the commitment cmt from the signature (consisting of challenge ch and response rsp) via Cmt.R and then checking if recomputing the challenge via the random oracle yields the correct challenge. For FS_{tr} , we could argue that each random oracle query, conditioned on there being no commitment collisions, has exactly one valid target. The same argument is not directly possible for FS_{ch} . The reason is that the commitments are not part of the signature but instead are recomputed using the commitment-recovery algorithm. Assume that adversary $\mathcal{A}$ is able to find a public key pk for which Cmt.R outputs the same commitment for multiple, say, t , signatures. Then making a random oracle query on said public key and (recovered) commitment together with an arbitrary message, suddenly has multiple valid targets: namely the t signatures for which the output of Cmt.R resulted in that commitment.

To circumvent this problem, we introduce a new security game $t\text{-Cmt-CR}_r$, which is shown in Fig. 4. This game grants an adversary access to a signing oracle and asks the adversary to find a public key which results in a multi-commitment collision of the commitment-recovery algorithm Cmt.R . We parameterize the game using an integer t , that specifies how large a collision has to be, as the hardness can vary depending on the value t .¹² Below we define the advantage of an adversary in winning $t\text{-Cmt-CR}_r$.

Definition 9. Let $\Sigma = (\text{KeyGen}, \text{Cmt.G}, \text{Rsp.G}, \text{Rsp.V}, \text{Cmt.R})$ be an identification scheme, $\mathcal{H}$ be a random oracle, and $\text{FS}_{\text{ch}}[\Sigma, \mathcal{H}]$ be the resulting Fiat-Shamir signature scheme. Let further the game $t\text{-Cmt-CR}_r$ be defined as in Fig. 4. For any adversary $\mathcal{A}$, making $q_S \geq t$ signing queries, its $t\text{-Cmt-CR}_r$ advantage is defined as

$$\text{Adv}_{\text{FS}_{\text{ch}}[\Sigma, \mathcal{H}]}^{t\text{-Cmt-CR}_r}(\mathcal{A}) := \Pr[t\text{-Cmt-CR}_r(\mathcal{A}) \rightarrow 1].$$

Using this new security game, we can show S-UEO security of FS_{ch} , provided that it is hard to find t -commitment collisions. An interesting aspect here is the role of t : under the condition that $\mathcal{A}$ cannot find a t -commitment collision, we know that each random oracle query has at most $t - 1$ targets (generously to the advantage

¹²Looking ahead, for Schnorr signatures, finding 2-collisions is trivial while larger collisions, already starting with 3-collisions, are significantly harder to find. For ML-DSA, LESS, SQISIGN, and CSI-FiSh finding 2-collisions is already hard.

Game $t\text{-Cmt-CR}_r$	Oracle $\text{Sign}(\text{sk}, \text{msg})$
$\mathcal{Q} \leftarrow \emptyset$ $(\text{pk}, \text{sk}) \leftarrow \text{\$KeyGen}()$ $(\overline{\text{pk}}, (\text{ch}_i, \text{rsp}_i)_{i \in [t]}) \leftarrow \mathcal{A}^{\text{Sign}(\text{sk}, \cdot)}(\text{pk})$ for $i \in [t]$ // Recompute commitments $\text{cmt}_i \leftarrow \text{Cmt.R}(\overline{\text{pk}}, \text{ch}_i, \text{rsp}_i)$ if $\overline{\text{pk}} = \text{pk}$ $\quad \textbf{return } 0$ // Key output is invalid if $\exists i \in [t]$ s.t. $(\text{ch}_i, \text{rsp}_i) \notin \mathcal{Q}$ $\quad \textbf{return } 0$ // Signature output is invalid if $\exists i \neq j$ s.t. $\text{cmt}_i \neq \text{cmt}_j$ $\quad \textbf{return } 0$ // Output is not a t -collision return 1	$(\text{ch}, \text{rsp}) \leftarrow \text{FS}_{\text{ch}}[\Sigma, \text{H}].\text{Sign}(\text{sk}, \text{msg})$ $\text{cmt} \leftarrow \text{\\$Cmt.G}(\text{sk})$ $\text{ch} \leftarrow \text{H}(\text{pk}, \text{cmt}, \text{msg})$ $\text{rsp} \leftarrow \text{Rsp.G}(\text{sk}, \text{cmt}, \text{ch})$ $\mathcal{Q} \leftarrow \mathcal{Q} \cup \{(\text{ch}, \text{rsp})\}$ return (ch, rsp)

Fig. 4: Security game $t\text{-Cmt-CR}_r$ for a signature scheme $\text{FS}_{\text{ch}}[\Sigma, \text{H}]$ following the Fiat-Shamir transform (challenge variant). The subscript r indicates “restricted” in the sense that the adversary has to find the commitment collision for signatures provided by the signing oracle Sign .

of the adversary assuming that it always finds a $(t - 1)$ -commitment collision). This means that our bound $\frac{q_H}{2^{|\text{ch}|}}$ obtains $(t - 1)$ as an additional factor. Hence, given that, say, 5-commitment collisions are hard to find, our bound increases by a factor of 4 which is equivalent to losing only 2 bits of security.

The theorem below establishes S-UEO security of FS_{ch} by reducing it to the restricted-multi-commitment-collision security of the signature scheme.

Theorem 10 (S-UEO Security of FS_{ch}). *Let Σ be an identification protocol and H be a random oracle. Consider the signature scheme $\text{FS}_{\text{ch}}[\Sigma, \text{H}]$. Then for any S-UEO adversary $\mathcal{A}$, making q_S queries to the signing oracle and q_H queries to H , there exists an adversary $\mathcal{B}$, making the same number of queries, such that, for $t \in [q_S]$, we have*

$$\text{Adv}_{\text{FS}_{\text{ch}}[\Sigma, \text{H}]}^{\text{S-UEO}}(\mathcal{A}) \leq \text{Adv}_{\text{FS}_{\text{ch}}[\Sigma, \text{H}]}^{t\text{-Cmt-CR}_r}(\mathcal{B}) + \frac{q_H(t - 1)}{2^{|\text{ch}|}}.$$

Proof. Let $\mathcal{A}$ be an adversary playing S-UEO. $\mathcal{A}$ is given a public key pk and access to the corresponding signing oracle $\text{Sign}(\text{sk}, \cdot)$. Denote by $(\text{msg}_1, \text{sig}_1 = (\text{ch}_1, \text{rsp}_1)), \dots, (\text{msg}_{q_S}, \text{sig}_{q_S} = (\text{ch}_{q_S}, \text{rsp}_{q_S}))$ the q_S message-signature pairs stemming from $\mathcal{A}$ ’s queries to Sign . In order to win S-UEO, the adversary has to find a message $\overline{\text{msg}}$ (not necessarily different from the queried ones) and a public key $\overline{\text{pk}} \neq \text{pk}$ such that one of the signatures $\text{sig}_1, \dots, \text{sig}_{q_S}$ verifies $\overline{\text{msg}}$ under $\overline{\text{pk}}$. By definition of the verification algorithm for FS_{ch} schemes, this implies that $\text{ch}_i = \text{H}(\overline{\text{pk}}, \text{Cmt.R}(\overline{\text{pk}}, \text{ch}_i, \text{rsp}_i), \overline{\text{msg}})$ has to hold for some i .

Denote by $\mathcal{PK}$ the set of public keys, the adversary uses for the queries to the random oracle H . For $\mathbf{pk}' \in \mathcal{PK}$, we define $\mathcal{CMT}(\mathbf{pk}') = \{\text{Cmt.R}(\mathbf{pk}', \mathbf{ch}_i, \mathbf{rsp}_i) \mid i \in \{1, \dots, q_S\}\}$ and call a query $(\mathbf{pk}', \mathbf{cmt}, \mathbf{msg})$ of $\mathcal{A}$ to the random oracle H *relevant*, if $\mathbf{cmt} \in \mathcal{CMT}(\mathbf{pk}')$. Note that $\mathcal{A}$ can only win with a relevant query, as otherwise the verification check will not be successful. Hence, we assume that the adversary makes only relevant queries in the following.

For $t \in \{2, \dots, q_S\}$, let $\mathbf{tColl}$ be the event that there is $\mathbf{pk}' \in \mathcal{PK}$ such that at least t of the commitments $\mathbf{cmt}_i = \text{Cmt.R}(\mathbf{pk}', \mathbf{ch}_i, \mathbf{rsp}_i)$ for $i \in \{1, \dots, q_S\}$ agree. Then we obtain

$$\begin{aligned} \mathbf{Adv}_{\text{FS}_{\text{ch}}[\Sigma, H]}^{\text{S-UEO}}(\mathcal{A}) &= \Pr[\mathcal{A} \text{ wins S-UEO}] \\ &\leq \Pr[\mathbf{tColl}] + \Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg \mathbf{tColl}]. \end{aligned}$$

To bound the second summand, i.e., $\Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg \mathbf{tColl}]$, we note that conditioned on $\neg \mathbf{tColl}$, the best query to H can have at most $(t-1)$ targets (namely if $t-1$ of the commitments collide and all $t-1$ corresponding challenges are different). Then, for each random oracle query, the adversary has a chance of at most $\frac{(t-1)}{2^{|\text{ch}|}}$. This yields

$$\Pr[\mathcal{A} \text{ wins S-UEO} \mid \neg \mathbf{tColl}] \leq \frac{q_H(t-1)}{2^{|\text{ch}|}}.$$

We now show that the first summand, i.e., $\Pr[\mathbf{tColl}]$, can be bounded by the t -Cmt-CR $_r$ security of the signature, using the game t -Cmt-CR $_r$ described in Fig. 4. We give a straightforward reduction from an adversary $\mathcal{A}$ that triggers the event $\mathbf{tColl}$ when playing S-UEO to an adversary $\mathcal{B}$ that wins the game t -Cmt-CR $_r$. Adversary $\mathcal{B}$ simply runs $\mathcal{A}$ on the same public key it receives as input and simulates the signing oracle by using its own signing oracle from t -Cmt-CR $_r$. As above we denote the message-signature pairs stemming from $\mathcal{A}$'s queries to the signing oracle by $(\mathbf{msg}_1, \mathbf{sig}_1 = (\mathbf{ch}_1, \mathbf{rsp}_1)), \dots, (\mathbf{msg}_{q_S}, \mathbf{sig}_{q_S} = (\mathbf{ch}_{q_S}, \mathbf{rsp}_{q_S}))$.

Assume $\mathcal{A}$ triggers the event $\mathbf{tColl}$, i.e., there is $\mathbf{pk}' \in \mathcal{PK}$ such that t of the commitments in $\{\text{Cmt.R}(\mathbf{pk}', \mathbf{ch}_i, \mathbf{rsp}_i) \mid i \in \{1, \dots, q_S\}\}$ agree. As $\mathcal{B}$ can monitor the random oracle queries made by $\mathcal{A}$, it can determine the public key $\mathbf{pk}'$ that yields the t -collision by testing all queried public keys together with the signatures resulting from $\mathcal{A}$'s signing queries. More precisely, for every queried public key $\mathbf{pk}'$, $\mathcal{B}$ locally computes $\text{Cmt.R}(\mathbf{pk}', \mathbf{ch}_i, \mathbf{rsp}_i)$, for $i \in \{1, \dots, q_S\}$, and checks for a t -commitment collision.

After re-indexing, we can assume wlog that the commitments stemming from the first t message-signature pairs collide for $\mathbf{pk}'$. Then $\mathcal{B}$ wins the game t -Cmt-CR $_r$ by outputting $\mathbf{pk}'$ and $\{\overline{\mathbf{msg}}_i, \overline{\mathbf{sig}}_i \mid i \in \{1, \dots, t\}\}$, as all of the signatures resulted from queries to the signing oracle and the commitments $\text{Cmt.R}(\mathbf{pk}', \mathbf{ch}_i, \mathbf{rsp}_i)$ agree for all $i \in \{1, \dots, t\}$. This finishes the proof. $\square$

Similar to the other variants, one might ask what effect weak keys might have on FS_{ch} . The theorem below states an attack against FS_{ch} , based on the identification scheme with weak keys Σ_w (cf. Fig. 3). This scheme has a weak key $\mathbf{pk}_w$ in the

sense that $\text{Cmt.R}(\text{pk}_w, \cdot, \cdot) = \text{cmt}_w$. Note that this attack does not contradict the bound given in [Theorem 10](#): for this scheme, finding colliding commitments is trivial, hence $\text{Adv}_{\text{FS}_{\text{ch}}[\Sigma_w, \text{H}]}^{t\text{-Cmt-CR}_r}(\mathcal{A}) = 1$ which makes the bound in [Theorem 10](#) trivial. The theorem below gives an attack against FS_{ch} using the weak identification scheme as shown in [Fig. 3](#).

Theorem 11 (S-UEO Attack against FS_{ch} with weak keys). *Let Σ_w be the identification scheme shown in [Fig. 3](#) and H be a random oracle. Consider the signature scheme $\text{FS}_{\text{ch}}[\Sigma_w, \text{H}]$ (following the description in [Fig. 1](#)). Then there exists an S-UEO adversary $\mathcal{A}$, making q_H and q_S queries to H and Sign , respectively, such that*

$$\text{Adv}_{\text{FS}_{\text{ch}}[\Sigma_w, \text{H}]}^{\text{S-UEO}}(\mathcal{A}) = \frac{q_S q_H}{2^{|\text{ch}|}}.$$

Proof. We construct an S-UEO adversary $\mathcal{A}$ as follows: given a public key pk , $\mathcal{A}$ queries q_S messages to the corresponding oracle $\text{Sign}(\text{sk}, \cdot)$, resulting in $\text{sig}_1 = (\text{ch}_1, \text{rsp}_1), \dots, \text{sig}_{q_S} = (\text{ch}_{q_S}, \text{rsp}_{q_S})$. In order to win S-UEO, the adversary has to find a public key $\bar{\text{pk}} \neq \text{pk}$ and a $\bar{\text{msg}}$ (not necessarily different from the queried messages $\text{msg}_1, \dots, \text{msg}_{q_S}$), such that there is $i^* \in \{1, \dots, q_S\}$ with $\text{Verify}(\bar{\text{pk}}, \bar{\text{msg}}, \text{sig}_{i^*}) = 1$. When choosing $\bar{\text{pk}} = \text{pk}_w$ and an arbitrary message $\bar{\text{msg}}$, the query $(\text{pk}_w, \text{cmt}_w, \bar{\text{msg}})$ to the random oracle H has q_S valid targets, namely $\text{ch}_1, \dots, \text{ch}_{q_S}$. This is the case as by construction $\text{Cmt.R}(\text{pk}_w, \text{ch}_i, \text{rsp}_i) = \text{cmt}_w$ holds for all $i \in \{1, \dots, q_S\}$. Repeating this for $q_H - 1$ further messages, the adversary wins with probability $\frac{q_S q_H}{2^{|\text{ch}|}}$. $\square$

Note that the effect of weak keys on the S-UEO security of FS_{ch} is quite different than on FS_{ct} . For FS_{ct} , we gave an attack (cf. [Theorem 7](#)) that had essentially no cost, i.e., the minimum of a single signing query. For FS_{ch} , our attack from [Theorem 11](#) requires more queries, namely, for $q_S = 2^{64}$ and $\lambda = 128$, it requires about 2^{64} random oracle queries.

3.5 Deterministic Fiat-Shamir (Challenge Variant)

We finally consider the deterministic version of Fiat-Shamir in the challenge variant. For this version, it turns out that the bounds on exclusive ownership are quite different. Looking at [Theorem 10](#), the best bound we can achieve is $\frac{q_H}{2^{|\text{ch}|}}$ —when finding 2-commitment collisions is impossible, i.e., $\text{Adv}^{2\text{-Cmt-CR}_r}(\mathcal{A}) = 0$. However, for neither of the schemes we are considering in this work, this is the case. What we can show though is that dFS_{ch} actually achieves the bound $\frac{q_H}{2^{|\text{ch}|}}$ when restricting to S-CEO. That is, the adversary has to find a different public key that verifies a message-signature pair obtained from the signing oracle. Since now the adversary only ever gets to see one signature per message, the message (which is input to H) effectively restricts the number of valid targets to 1: the challenge contained in the signature of that message. Even if the adversary can find (multi-)commitment collisions, these no longer increase the chance of the adversary in winning S-CEO. This is formalized in the theorem below.

$\text{dFS}_{\text{ch}}.\text{Sign}$	$\text{dFS}_{\text{ch}}.\text{Verify}$
$\text{cmt} \leftarrow \text{Cmt.G}(\text{sk}; \text{F}(\text{k}, \text{msg}))$	$(\text{ch}, \text{rsp}) \leftarrow \text{sig}$
$\text{ch} \leftarrow \text{H}(\text{pk}, \text{cmt}, \text{msg})$	$\text{cmt} \leftarrow \text{Cmt.R}(\text{pk}, \text{ch}, \text{rsp})$
$\text{rsp} \leftarrow \text{Rsp.G}(\text{sk}, \text{cmt}, \text{ch})$	if $\text{cmt} = \perp$
$\text{sig} \leftarrow (\text{ch}, \text{rsp})$	return 0
return sig	if $\text{ch} \neq \text{H}(\text{pk}, \text{cmt}, \text{msg})$
	return 0
	return 1

Fig. 5: Fiat-Shamir variant dFS_{ch} using a pseudorandom function F .

Theorem 12. *Let Σ be an identification protocol and H be a random oracle. Consider the signature scheme $\text{dFS}_{\text{ch}}[\Sigma, \text{H}]$ in Fig. 5. Then for any S-CEO adversary $\mathcal{A}$ making q_H queries to H , we have*

$$\text{Adv}_{\text{dFS}_{\text{ch}}[\Sigma, \text{H}]}^{\text{S-CEO}}(\mathcal{A}) \leq \frac{q_H}{2^{|\text{ch}|}}.$$

Proof. Let $\mathcal{A}$ be an S-CEO adversary against $\text{dFS}_{\text{ch}}[\Sigma, \text{H}]$, i.e., $\mathcal{A}$ is given a public key pk and access to the corresponding signing oracle $\text{Sign}(\text{sk}, \cdot)$. Denote by $(\text{msg}_1, \text{sig}_1 = (\text{ch}_1, \text{rsp}_1)), \dots, (\text{msg}_{q_S}, \text{sig}_{q_S} = (\text{ch}_{q_S}, \text{rsp}_{q_S}))$ the q_S message-signature pairs stemming from $\mathcal{A}$'s queries to Sign . In order to win S-CEO, the adversary has to find a public key $\text{pk} \neq \text{pk}$ which verifies one of the message-signature pairs resulting from the queries to Sign . Note that each of $\mathcal{A}$'s query $(\text{pk}, \text{cmt}, \text{msg}_i)$ to the random oracle H has *exactly* one target, namely the challenge ch_i corresponding to the queried message.¹³ In total, this implies that the probability of $\mathcal{A}$ winning S-CEO is upper bounded by $\frac{q_H}{2^{|\text{ch}|}}$. $\square$

Note that the reasoning does not carry over to S-DEO security (and hence S-UEO). Since the adversary has to output a *different* message than the one it used in the signing query, a random oracle query on a message that was never queried to the sign oracle can hit any challenge. Thus, multi-commitment collisions again increase the probability of each random oracle query by the adversary.

4 Analysis of Real-World Signature Schemes

Having established the general results in the previous section, we now turn towards specific signature schemes. As mentioned, the most commonly used variant of Fiat-Shamir is FS_{ch} , which is also used by all signature schemes we are analyzing here. Looking at Theorem 10, it remains to analyze the hardness of finding

¹³For a non-deterministic scheme, it would be possible to choose the same message for multiple queries to the sign oracle, while still obtaining different challenges with high probability—for the deterministic scheme $\text{dFS}_{\text{ch}}[\Sigma, \text{H}]$, this is not possible.

KeyGen()	Sign(sk, msg)	Verify(pk, msg, sig)	Cmt.R(pk, sig)
$z \leftarrow \$_{[q-1]}$	$z \leftarrow \text{sk}$	$(e, s) \leftarrow \text{sig}$	$Z \leftarrow \text{pk}$
$Z \leftarrow g^z$	$k \leftarrow \$_{[q-1]}$	$r \leftarrow \text{Cmt.R}(\text{pk}, \text{sig})$	$(e, s) \leftarrow \text{sig}$
$\text{sk} \leftarrow z$	$r \leftarrow g^k$	if $H(Z, \text{msg}, r) \neq e$	$r \leftarrow g^s Z^{-e}$
$\text{pk} \leftarrow Z$	$e \leftarrow H(Z, \text{msg}, r)$	return 0	return r
return (sk, pk)	$s \leftarrow k + ze$	return 1	
	$\text{sig} \leftarrow (e, s)$		
	return sig		

Fig. 6: Fiat-Shamir signature scheme SCHNORR (including key prefixing) with an explicit description of the commitment recovery algorithm **Cmt.R**. The values r , e , and s correspond to **cmt**, **ch**, and **rsp**, respectively.

multi-commitment collisions as this provides a bound on the S-UEO security of the signature scheme.

In Section 4.1, we analyze the Schnorr signature scheme while ML-DSA is covered in Section 4.2. The NIST round-2 candidates SQIsign and LESS are analyzed in Section 4.3 and Section 4.4, respectively. Appendix A.1 considers the CSI-FiSh signature scheme.

4.1 The DL-Based Signature Scheme by Schnorr

Description. Schnorr developed a signature scheme over cyclic groups whose security is based on the hardness of the discrete logarithm (DL) problem [Sch91]. The signature scheme is constructed by applying (the challenge variant of) Fiat-Shamir to an underlying identification scheme. A description of the scheme, using key prefixing, is given in Fig. 6. Schnorr signatures work on a cyclic group of prime order q ($\geq 2^\lambda$, for security parameter λ), generated by a generator g .

Analysis. It turns out that finding a 2-commitment collisions is easy for Schnorr signatures. Since the commitments are recovered by computing $g^{s_i} Z^{e_i}$, one simply sets these two commitments to be equal and solves for the public key Z . This is formalized in the theorem below.

Theorem 13. *Let SCHNORR be the signature scheme as shown in Fig. 6. There exists an adversary $\mathcal{A}$ such that*

$$\text{Adv}_{\text{SCHNORR}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) = 1.$$

Proof. Adversary $\mathcal{A}$ makes two signing queries (on arbitrary messages msg_1 and msg_2), which result in signatures $\text{sig}_1 = (e_1, s_1)$ and $\text{sig}_2 = (e_2, s_2)$. Then $\mathcal{A}$ computes $\overline{\text{sk}} \leftarrow (s_1 - s_2)(e_1 - e_2)^{-1} \bmod q$, $\overline{\text{pk}} \leftarrow g^{\overline{\text{sk}}}$, and outputs $\overline{\text{pk}}$.

It holds that

$$\text{Cmt.R}(\overline{\text{pk}}, e_1, s_1) = g^{s_1} Z^{-e_1} = g^{\frac{e_1 s_2 - e_2 s_1}{(e_1 - e_2)}} = g^{s_2} Z^{-e_2} = \text{Cmt.R}(\overline{\text{pk}}, e_2, s_2),$$

i.e., the outputs of Cmt.R for the two signatures collide for the public key $\overline{\text{pk}}$. $\square$

The above shows that we cannot leverage [Theorem 10](#) for $t = 2$ since the bound on the S-UEO advantage is larger than 1 in this case. However, increasing t only slightly already has significant effect on the t -Cmt-CR $_r$ advantage as shown in the following theorem.

Theorem 14. *Let SCHNORR be the signature scheme as shown in [Fig. 6](#). Then for any adversary $\mathcal{A}$, making q_S signing queries, and for $t \in [q_S]$, it holds that*

$$\text{Adv}_{\text{SCHNORR}}^{t\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{t} \frac{1}{2^{|\text{ch}|(t-2)}}.$$

Proof. Let $\text{sig}_1 = (e_1, s_1), \dots, \text{sig}_{q_S} = (e_{q_S}, s_{q_S})$ be the q_S signatures that $\mathcal{A}$ receives from its oracle. Using the proof of [Theorem 13](#), for every pair of signatures $\text{sig}_i \neq \text{sig}_j$, there is exactly one public key ($Z = g^{(s_1-s_2)(e_1-e_2)^{-1}}$) which leads to a collision of Cmt.R for these two signatures. If we consider a 3-commitment collision (for signatures sig_i , sig_j , and sig_k), it holds that $Z_{ij} = g^{(s_i-s_j)(e_i-e_j)^{-1}}$ results in $\text{Cmt.R}(Z_{ij}, e_i, s_i) = \text{Cmt.R}(Z_{ij}, e_j, s_j)$, while $Z_{ik} = g^{(s_i-s_k)(e_i-e_k)^{-1}}$ results in $\text{Cmt.R}(Z_{ik}, e_i, s_i) = \text{Cmt.R}(Z_{ik}, e_k, s_k)$. In order to have a 3-commitment collision, we thus need $Z_{ij} = Z_{ik}$ which translates to $(s_i - s_j)(e_i - e_j)^{-1} = (s_i - s_k)(e_i - e_k)^{-1}$. This is equivalent to

$$e_j = e_i - (s_i - s_j)(s_i - s_k)^{-1}(e_i - e_k). \quad (1)$$

By construction, the output of the random oracle is uniformly distributed. Therefore, [Eq. \(1\)](#) is true with probability $\frac{1}{2^{|\text{ch}|}}$.

Extending the above argument to t -commitment collision yields a probability of $\frac{1}{2^{|\text{ch}|(t-2)}}$ that there exists a public key pk such that t commitments (out of $\text{Cmt.R}(\text{pk}, e_1, s_1), \dots, \text{Cmt.R}(\text{pk}, e_{q_S}, s_{q_S})$) collide under this key.

To overall bound the probability that $\mathcal{A}$ can find a t -commitment collision, we need to consider all possible choices of selecting t signatures out of the q_S signatures. This results in the following bound

$$\text{Adv}_{\text{SCHNORR}}^{t\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{t} \frac{1}{2^{|\text{ch}|(t-2)}},$$

where the first factor covers all combinations of t signatures that can collide while the second factor bounds the probability for each tuple of t signatures to collide. This concludes the proof. $\square$

The corollary below provides a bound on the S-UEO advantage of an adversary against the Schnorr signature scheme. It follows directly from [Theorem 10](#) and [Theorem 14](#).

Corollary 15. *Let SCHNORR be the signature scheme as shown in [Fig. 6](#). Then for any adversary $\mathcal{A}$, making q_S and q_H signing queries and random oracle queries, respectively, and for $t \in [q_S]$, it holds that*

$$\text{Adv}_{\text{SCHNORR}}^{\text{S-UEO}}(\mathcal{A}) \leq \binom{q_S}{t} \frac{1}{2^{|\text{ch}|(t-2)}} + \frac{(t-1)q_H}{2^{|\text{ch}|}}.$$

Table 1: Upper bound on the S-UEO advantage for Schnorr signatures for different values of t (multi-commitment collision) and $q_S = 2^{64}$ (signing queries). As can be seen from the table, for $t = 3$, the bound is trivial. For $t = 4$ and $t = 5$, the bounds are no longer trivial but still insufficient. However, for $t = 6$, the bound becomes significant as the term for the multi-commitment collision becomes smaller than $2^{-\lambda}$.

λ	t	q_S	$\mathbf{Adv}_{\text{SCHNORR}}^{\text{S-UEO}}(\mathcal{A})$
128	3	2^{64}	$\leq 2^{62} + \frac{2q_H}{2^{ \text{ch} }}$
128	4	2^{64}	$\leq \frac{1}{2^4} + \frac{3q_H}{2^{ \text{ch} }}$
128	5	2^{64}	$\leq \frac{1}{2^{70}} + \frac{4q_H}{2^{ \text{ch} }}$
128	6	2^{64}	$\leq \frac{1}{2^{137}} + \frac{5q_H}{2^{ \text{ch} }}$

Corollary 15 gives a bound on the S-UEO security of the Schnorr signature scheme in terms of a parameter t . Specific bounds, for varying values of t and signing queries q_S , are shown in Table 1. Note that these are all upper bounds and we simply take the minimum for S-UEO security. This is the case for $t = 6$, where we obtain about 125-bit S-UEO security.

4.2 The Lattice-Based Signature Scheme ML-DSA

Description. ML-DSA [NIST24a] is a module-lattice-based signature scheme that was standardized by NIST as a result of their PQC standardization process [NIST17]. It is based on the scheme Dilithium [LDK+20], which was a submission to the standardization process. ML-DSA applies the Fiat-Shamir transform in the challenge variant with key prefixing.¹⁴ In the following, we give only the details on ML-DSA that are relevant for our analysis, and refer to the specification [NIST24a] for more information.¹⁵

For a prime q , consider the ring $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$. Further, let $k, l > 0$ be integers and denote by $\mathbb{B}$ the set $\{0, 1, \dots, 255\}$. An ML-DSA public key consists of a public seed ρ and $t \in R_q^k$. The latter is a compressed version (using the algorithm **Power2Round**) of an LWE sample $As_1 + s_2$ for $A \in R_q^{k \times l}$ a random matrix expanded from ρ and $s_1 \in R_q^l, s_2 \in R_q^k$ two random short vectors. Note that ML-DSA switches between presentations using the NTT as an isomorphism between R_q and $\mathbb{Z}_q^{256}$. For sake of simplicity, we do not explicitly

¹⁴Note that in ML-DSA, the public key and message are first hashed, and the resulting *message representative* is then used for the challenge computation, whereas in our definition of key prefixing, public key and message are directly used.

¹⁵For instance, we drop the encoding function used to encode the final signature.

Cmt.R(pk, sig)	UseHint(h, r)
$(\rho, t) \leftarrow \mathbf{pk}$	$m \leftarrow \frac{q-1}{2\gamma_2} \quad // \gamma_2 \text{ is the low-order rounding range}$
$A \leftarrow \mathbf{ExpandA}(\rho)$	$(r_1, r_0) \leftarrow \mathbf{Decompose}(r)$
$(\tilde{c}, z, h) \leftarrow \mathbf{sig}$	if $h = 1$ and $r_0 > 0$ return $(r_1 + 1) \bmod m$
$c \leftarrow \mathbf{SampleInBall}(\tilde{c})$	if $h = 1$ and $r_0 \leq 0$ return $(r_1 - 1) \bmod m$
$w' \leftarrow \mathbf{UseHint}(h, Az - ct2^d)$	return r_1
$// \mathbf{UseHint}$ is applied component-wise	
return w'	

Fig. 7: Commitment recovery algorithm **Cmt.R** of ML-DSA. Note that **SampleInBall** is used to pseudorandomly generate a polynomial $c \in R_q$ and **Decompose** divides $r \in \mathbb{Z}_q$ into high bits r_1 and low bits r_0 .

write down applications of NTT and NTT^{-1} . An ML-DSA signature is of the form $\mathbf{sig} = (\tilde{c}, z, h)$, where $\tilde{c} \in \mathbb{B}^{\lambda/4}$ is the challenge,¹⁶ $z \in R_q^l$ the response, and $h \in R_2^k$ a hint used to recover the commitment during verification. The commitment recovery algorithm is described in Fig. 7. On input $(\mathbf{pk}, \mathbf{sig})$, the algorithm outputs a vector $w' \in \mathbb{Z}^k$ with entries reduced modulo $m = \frac{q-1}{2\gamma_2}$ for γ_2 the low-order rounding range.

Analysis. In order to apply Theorem 10, we analyze ML-DSA with respect to the game $t\text{-Cmt-CR}_r$. The following theorem shows that it is already hard to break $t\text{-Cmt-CR}_r$ for $t = 2$. This can be traced back mainly to two properties: Firstly, the adversary cannot choose the public key matrix A freely, but only a seed from which the matrix is then pseudorandomly expanded from. Secondly, for a challenge $\tilde{c}$, only a small number of the coefficients of $\mathbf{SampleInBall}(\tilde{c})$ are not 0. Due to this, more than half of the equations that have to be solved, do not depend on the public key vector t . In combination, these two points imply that the probability of finding a collision of two commitments will reduce to whether a number of randomly sampled bits agree, hence yielding a good bound.

Theorem 16. *Let ML-DSA be the signature scheme as described above with **ExpandA** modeled as a random oracle. Then for any adversary $\mathcal{A}$, making q_S signing queries and q_H random oracle queries, it holds that*

$$\mathbf{Adv}_{\text{ML-DSA}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{2} \frac{q_H}{2^{h(256-2\tau)}},$$

where h is the number of high bits and τ is the (fixed) Hamming weight of $\mathbf{SampleInBall}(\tilde{c})$ for a challenge $\tilde{c}$.

Proof. Let $\mathcal{A}$ be an adversary playing the game 2-Cmt-CR_r , i.e., it is given $\mathbf{pk}$ and access to the corresponding sign oracle $\mathbf{Sign}(\mathbf{sk}, \cdot)$ and outputs $(\bar{\mathbf{pk}}, \mathbf{sig}_1 =$

¹⁶Note that in the ML-DSA specification, $\tilde{c}$ is referred to as *commitment hash* and only after applying **SampleInBall**, the result is called challenge.

$(\tilde{c}_1, z_1, h_1), \mathbf{sig}_2 = (\tilde{c}_2, z_2, h_2))$. In order to win, the signatures $\mathbf{sig}_1$ and $\mathbf{sig}_2$ must stem from queries to the sign oracle and it has to hold that $\overline{\mathbf{pk}} \neq \mathbf{pk}$ and $\mathbf{cmt}_1 := \mathbf{Cmt.R}(\overline{\mathbf{pk}}, \tilde{c}_1, z_1, h_1) = \mathbf{Cmt.R}(\overline{\mathbf{pk}}, \tilde{c}_2, z_2, h_2) =: \mathbf{cmt}_2$.

To begin, we assume the adversary makes only two sign queries resulting in $\mathbf{sig}_1$ and $\mathbf{sig}_2$. By definition of the commitment recovery algorithm, the recomputed commitments agree if and only if

$$\mathbf{UseHint}(h_1, \overline{A}z_1 - c_1 \bar{t}2^d) = \mathbf{UseHint}(h_2, \overline{A}z_2 - c_2 \bar{t}2^d), \quad (2)$$

where $\overline{\mathbf{pk}} = (\bar{\rho}, \bar{t})$, $\overline{A} = \mathbf{ExpandA}(\bar{\rho})$, and $c_i = \mathbf{SampleInBall}(\tilde{c}_i)$. The algorithm $\mathbf{UseHint}$ acts component-wise on $(\overline{A}z_i - c_i \bar{t}2^d)$; it takes the high bits, adds $\lambda_i \in \{-1, 0, 1\}$ (depending on the value of the hint h_i and $\overline{A}z_i - c_i \bar{t}2^d$), and reduces modulo m . Thus, Eq. (2) can be rewritten as follows, for some $\lambda \in \{-2, -1, 0, 1, 2\}^{256}$

$$\mathbf{HighBits}(\overline{A}z_1 - c_1 \bar{t}2^d) = \mathbf{HighBits}(\overline{A}z_2 - c_2 \bar{t}2^d) + \lambda \pmod{m}. \quad (3)$$

Note that $c_1, c_2 \in R_q$ have coefficients in $\{-1, 0, 1\}$ and Hamming weight τ . In particular, for c_1 and c_2 , there are at least $256 - 2\tau$ indices i for which $c_{1,i} = c_{2,i} = 0$. Denote the set containing all these indices by I . Then Eq. (3) implies that $\mathbf{HighBits}((\overline{A}z_1)_i) = \mathbf{HighBits}((\overline{A}z_2)_i) + \lambda_i \pmod{m}$ for all $i \in I$. Since $\overline{A} = \mathbf{ExpandA}(\bar{\rho})$ and $\mathbf{ExpandA}$ is assumed to be a random oracle, the adversary has no control over $\overline{A}$ by choice of $\bar{\rho}$. Moreover, the probability that the high bits of $(\overline{A}z_1)_i$ equal $\mathbf{HighBits}((\overline{A}z_2)_i) + \lambda_i$ modulo m is $\frac{1}{2}$ for each bit. Hence, for h the number of high bits, we obtain

$$\Pr[\mathbf{cmt}_1 = \mathbf{cmt}_2] \leq \Pr[\mathbf{cmt}_{1,i} = \mathbf{cmt}_{2,i} \mid i \in I] \leq \left(\frac{1}{2^h}\right)^{|I|} \leq \left(\frac{1}{2^h}\right)^{256-2\tau}.$$

Since the adversary can make q_H queries to the random oracle (i.e., try for q_H many matrices $\overline{A}$ whether the recomputed commitments agree), the probability to win 2-Cmt-CR_r with two sign queries is $p = \frac{q_H}{2^{h(256-2\tau)}}$.

Using this, we can generalize to the case that the adversary makes q_S queries to the sign oracle. For each pair of two signatures out of the q_S signatures resulting from the sign queries, the probability to win is p . Since there are $\binom{q_S}{2}$ possibilities to choose two signatures out of q_S , we obtain the final bound as

$$\mathbf{Adv}_{\text{ML-DSA}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{2} \frac{q_H}{2^{h(256-2\tau)}}. \quad \square$$

Combining the above result and the general S-UEO bound from Theorem 10, we obtain the following corollary for the S-UEO security of ML-DSA.

Corollary 17. *Let ML-DSA be the signature scheme as described in [NIST24a]. Then for any adversary $\mathcal{A}$, making q_S signing queries, $q_{H,1}$ random oracle queries to $\mathbf{ExpandA}$, and $q_{H,2}$ random oracle queries to $\mathbf{H}$ (random oracle from the Fiat-Shamir transform), it holds that*

$$\mathbf{Adv}_{\text{ML-DSA}}^{\text{S-UEO}}(\mathcal{A}) \leq \binom{q_S}{2} \frac{q_{H,1}}{2^{h(256-2\tau)}} + \frac{q_{H,2}}{2^{256}},$$

Table 2: Bounds for the S-UEO advantage of ML-DSA. Here, h is the number of high bits and τ is the (fixed) Hamming weight of $\text{SampleInBall}(\tilde{c})$ for a challenge $\tilde{c}$.

	λ	τ	h	$\mathbf{Adv}_{\text{ML-DSA}}^{\text{S-UEO}}(\mathcal{A})$
ML-DSA-44 (NIST Level II)	128	39	6	$\leq \binom{q_S}{2} \frac{q_H}{2^{1068}} + \frac{q_H}{2^{256}}$
ML-DSA-65 (NIST Level III)	192	49	5	$\leq \binom{q_S}{2} \frac{q_H}{2^{790}} + \frac{q_H}{2^{256}}$
ML-DSA-87 (NIST Level V)	256	60	5	$\leq \binom{q_S}{2} \frac{q_H}{2^{680}} + \frac{q_H}{2^{256}}$

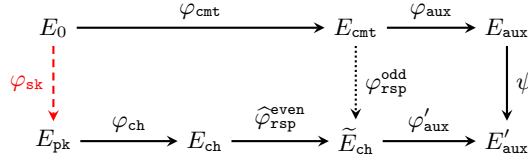


Fig. 8: Overview of the SQISign identification scheme.

where τ is the Hamming weight of $\text{SampleInBall}(\tilde{c})$ for a challenge $\tilde{c}$ and h is the number of high bits.

Plugging in the concrete ML-DSA parameters for the three security levels, we obtain the bounds shown in Table 2. Note that this significantly improves the S-UEO bound for ML-DSA, as prior results reduced the security to collision resistance of the hash function (i.e., 128 bits of security due to the challenge length being 256 bits).

4.3 The Isogeny-Based Signature Scheme SQISign

Description. SQISign is a signature scheme based on isogenies between supersingular elliptic curves [AAA⁺24]. It applies the Fiat-Shamir transform in the challenge variant with key prefixing to an isogeny-based identification scheme. For reasons of brevity, we will not give a full introduction on the background of SQISign, but refer to [AAA⁺24], in particular [AAA⁺24, Chapter 1] for a high-level description. Note that the round-2 submission of SQISign is based on SQISign2D-West [BDD⁺24] and therefore differs from the round-1 submission [CSD⁺23] in various aspects. Therefore, earlier analyses of BUFF features [ADM⁺24] targeting the round-1 submission of SQISign or variants like Ap $\grave{r}$ esSQI [CEMR24] do not apply.

We provide an overview of the SQISign identification scheme in Fig. 8.¹⁷ We work with supersingular elliptic curves over $\mathbb{F}_{p^2}$ with a prime p , and fix

¹⁷We slightly simplify the presentation, avoiding the notion of backtracking isogenies.

a starting curve E_0 . The prover's secret key is an isogeny $\varphi_{\text{sk}} : E_0 \rightarrow E_{\text{pk}}$ to the public key curve E_{pk} . The prover computes a random isogeny $\varphi_{\text{cmt}} : E_0 \rightarrow E_{\text{cmt}}$ and outputs (the j -invariant of) E_{cmt} as commitment. The verifier sends as challenge an isogeny $\varphi_{\text{ch}} : E_{\text{pk}} \rightarrow E_{\text{ch}}$ and the prover replies with an isogeny $\varphi_{\text{rsp}} : E_{\text{cmt}} \rightarrow E_{\text{ch}}$. For technical reasons, we split this isogeny in an odd-degree and 2-power-degree isogeny, i.e., $\varphi_{\text{rsp}} = \varphi_{\text{rsp}}^{\text{odd}} \circ \varphi_{\text{rsp}}^{\text{even}}$, such that $\tilde{\varphi}_{\text{rsp}}^{\text{even}} : E_{\text{ch}} \rightarrow \tilde{E}_{\text{ch}}$. This allows to define auxiliary curves E_{aux} and E'_{aux} such that they form a Kani diagram (see [CD23, BDD⁺24]) together with E_{cmt} and $\tilde{E}_{\text{ch}}$. Thus, the prover can represent $\varphi_{\text{rsp}}^{\text{odd}}$ efficiently through interpolation data. The verifier then recomputes $\varphi_{\text{rsp}}^{\text{odd}}$ through a 2-dimensional isogeny and verifies that indeed $\varphi_{\text{rsp}} : E_{\text{cmt}} \rightarrow E_{\text{ch}}$.

SQISIGN defines a prime $p = c \cdot 2^f - 1$ of $\approx 2\lambda$ bits and f as large as possible, i.e., $f \approx 2\lambda$. Furthermore, $e_{\text{rsp}} = \lceil \log_2(\sqrt{p}) \rceil$ serves as upper bound $2^{e_{\text{rsp}}}$ for the response isogeny degree, and $e_{\text{ch}} = f - e_{\text{rsp}} \approx \lambda$ defines the challenge space through $0 < \text{ch} < 2^{e_{\text{ch}}}$. The signer computes the challenge as $\text{ch} = \text{H}(\text{pk}, j(E_{\text{cmt}}), \text{msg})$, where $j(E)$ denotes the j -invariant of the curve E .

A public key pk consists of a tuple $(E_{\text{pk}}, \text{hint}_{\text{pk}})$, where E_{pk} is a supersingular elliptic curve and the integer hint_{pk} allows for efficiently generating a deterministic basis $(P_{\text{pk}}, Q_{\text{pk}})$ of the 2^f -torsion of E_{pk} , denoted as $E_{\text{pk}}[2^f]$.

A signature sig consists of $(E_{\text{aux}}, n_{\text{bt}}, r_{\text{rsp}}, \mathbf{M}_{\text{ch}}, \text{ch}, \text{hint}_{\text{aux}}, \text{hint}_{\text{ch}})$. We will explain the role of each of these components during a high-level explanation of the commitment recovery algorithm **Cmt.R** that is called during verification. In an initial step, the verification checks if $e'_{\text{rsp}} = e_{\text{rsp}} - n_{\text{bt}} - r_{\text{rsp}} \leq 0$ or if one of the curves E_{pk} or E_{aux} is not supersingular, in which case we reject the signature. Then, **Cmt.R** executes the following steps:

1. Computation of E_{ch} : From hint_{pk} , we compute a basis $(P_{\text{pk}}, Q_{\text{pk}})$ of $E_{\text{pk}}[2^f]$. From ch , we compute $P_{\text{pk}} + [\text{ch}]Q_{\text{pk}}$ from which we derive a kernel generator that defines the challenge isogeny $\varphi_{\text{ch}} : E_{\text{pk}} \rightarrow E_{\text{ch}}$. Using hint_{ch} , we compute a deterministic basis $(P_{\text{ch}}, Q_{\text{ch}})$ of $E_{\text{ch}}[2^f]$, and scale it by the matrix $\mathbf{M}_{\text{ch}}$.
2. Computation of $\tilde{E}_{\text{ch}}$: If $r_{\text{rsp}} > 0$, we compute a $2^{r_{\text{rsp}}}$ -isogeny $\tilde{\varphi}_{\text{rsp}}^{\text{even}} : E_{\text{ch}} \rightarrow \tilde{E}_{\text{ch}}$ defined by a point derived from P_{ch} resp. Q_{ch} . We evaluate $(P_{\text{ch}}, Q_{\text{ch}})$ under $\tilde{\varphi}_{\text{rsp}}^{\text{even}}$ and derive a basis $(\tilde{P}_{\text{ch}}, \tilde{Q}_{\text{ch}})$ of $\tilde{E}_{\text{ch}}[2^{e'_{\text{rsp}}+2}]$. If $r_{\text{rsp}} = 0$, we set $\tilde{E}_{\text{ch}} = E_{\text{ch}}$ and derive $(\tilde{P}_{\text{ch}}, \tilde{Q}_{\text{ch}})$ from $(P_{\text{ch}}, Q_{\text{ch}})$.
3. Computation of E_{cmt} : Making use of hint_{aux} , we compute a deterministic basis $(P_{\text{aux}}, Q_{\text{aux}})$ of $E_{\text{aux}}[2^{e'_{\text{rsp}}+2}]$. For honestly generated inputs, the tuples $(\tilde{E}_{\text{ch}}, \tilde{P}_{\text{ch}}, \tilde{Q}_{\text{ch}})$ and $(E_{\text{aux}}, P_{\text{aux}}, Q_{\text{aux}})$ give rise to a Kani diagram. Hence, we can compute a 2-dimensional $(2, 2)$ -isogeny $\Phi : \tilde{E}_{\text{ch}} \times E_{\text{aux}} \rightarrow E_{\text{cmt}} \times E'_{\text{aux}}$ between products of elliptic curves, therefore recovering the commitment curve E_{cmt} .

The final step of **Verify** then checks whether $\text{ch} = \text{H}(\text{pk}, j(E_{\text{cmt}}), \text{msg})$. If during the verification input points for isogenies do not satisfy the necessary requirements, such as having the correct order, the signature is rejected.

Analysis. We will argue that it is hard for an adversary to construct a 2-commitment collision in SQUISH for q_S given signatures. However, it appears difficult to rigorously prove bounds for the adversary's advantage. Hence, we rely on heuristic arguments and hardness assumptions that are backed by prior work [AAA⁺24, CD23, ADM⁺24]. A rigorous proof is beyond the scope of this work, and is left as future work.

Let sig_1 and sig_2 be honestly generated signatures. The adversary $\mathcal{A}$ tries to find a new public key $\overline{\text{pk}}$ such that **Cmt.R** recovers the same commitment curve E_{cmt} , resp. the same j -invariant, for sig_1 and sig_2 . To simplify the analysis, we assume to $\mathcal{A}$'s advantage that even when changing E_{pk} , all points occurring during **Cmt.R** satisfy all requirements, such as having the correct orders. This is not true in practice, since especially hint_{ch} may no longer fit to E_{ch} when using a different E_{pk} than expected.

First, we note that $\overline{\text{pk}}$ can be chosen to differ from the original pk only in the choice of hint_{pk} .¹⁸ Especially when choosing $\text{hint}_{\overline{\text{pk}}} = 0$ (assuming $\text{hint}_{\text{pk}} \neq 0$ such that $\text{pk} \neq \overline{\text{pk}}$), **Cmt.R** would proceed as when using pk . It would thus recover the original $E_{\text{cmt}}^{(1)}$ resp. $E_{\text{cmt}}^{(2)}$ used in the generation of sig_1 resp. sig_2 . However, the commitment-collision probability in this case is roughly $\frac{1}{p} \approx \frac{1}{2^{2\lambda}}$.

As described above, **Cmt.R** recovers E_{cmt} from a 2-dimensional isogeny Φ , requiring that $\tilde{E}_{\text{ch}}$ and E_{aux} together with the associated points give rise to a Kani diagram. However, $\tilde{E}_{\text{ch}}$ is computed in the first steps of **Cmt.R** depending on $\overline{\text{pk}}$, and the adversary is not able to directly choose it. As a first case, we assume that $\mathcal{A}$ has no direct control over the resulting $\tilde{E}_{\text{ch}}$. We therefore model it as a random curve. Then, with overwhelming probability $1 - 1/p$ the codomain of Φ is the Jacobian of a hyperelliptic curve, instead of a product of elliptic curves, since the involved curves do not form a Kani diagram (see [CD23, BDD⁺24]). Hence, the signature gets rejected, via **Cmt.R** outputting $\perp$. For the two given signatures, we obtain a probability of $1/p^2$ per choice of $\overline{\text{pk}}$ to recover elliptic curves $E_{\text{cmt}}^{(1)}$ resp. $E_{\text{cmt}}^{(2)}$ instead of **Cmt.R** returning $\perp$ in at least one case, while there are roughly p choices for $\overline{\text{pk}}$. This leads to a total probability of $1/p$ for $\mathcal{A}$ to be able to find a suitable $\overline{\text{pk}}$ to recover valid commitments. Note that this probability even ignores the fact that we require $j(E_{\text{cmt}}^{(1)}) = j(E_{\text{cmt}}^{(2)})$, which again has a probability of roughly $1/p$, lowering the total success probability for a commitment collision to $1/p^2$.

To circumvent this randomized approach, we assume as second case that for sig_1 and sig_2 , $\mathcal{A}$ knows curves $\tilde{E}_{\text{ch}}^{(1)}$ resp. $\tilde{E}_{\text{ch}}^{(2)}$ such that their occurrence implies that **Cmt.R** outputs the same commitment E_{cmt} (resp. equal j -invariants) for both signatures.¹⁹ Thus, $\mathcal{A}$ needs to find a public key $\overline{\text{pk}}$ such that **Cmt.R** produces the desired $\tilde{E}_{\text{ch}}^{(1)}$ for sig_1 resp. $\tilde{E}_{\text{ch}}^{(2)}$ for sig_2 . To bound the complexity for this, we assume the easiest case for the adversary, i.e., $r_{\text{rsp}}^{(1)} = r_{\text{rsp}}^{(2)} = 0$, and $\varphi_{\text{ch}}^{(1)}$ as well as

¹⁸Note that hint_{pk} is an input to the random oracle to compute **ch**, and therefore changing hint_{pk} also changes **ch**, preventing a simple S-CEO attack.

¹⁹Note that for the given restrictions imposed by the verification in SQUISH, it is not clear whether such a curve exists in the first place.

Table 3: Bounds for the S-UEO advantage of the SQISIGN parameter sets.

	λ	$\mathbf{Adv}_{\text{SQISIGN}}^{\text{S-UEO}}(\mathcal{A})$
SQISIGN (NIST Level I)	128	$\leq \binom{q_S}{2} \frac{1}{2^{256}} + \frac{q_H}{2^{128}}$
SQISIGN (NIST Level III)	192	$\leq \binom{q_S}{2} \frac{1}{2^{384}} + \frac{q_H}{2^{192}}$
SQISIGN (NIST Level V)	256	$\leq \binom{q_S}{2} \frac{1}{2^{512}} + \frac{q_H}{2^{256}}$

$\varphi_{\text{ch}}^{(2)}$ being of degree $2^{e_{\text{ch}}} \approx 2^\lambda$ for both signatures. This problem is similar to the analysis in [ADM⁺24] that involved solving this for a single signature instead of a pair of signatures. For a single signature, [ADM⁺24] proposes a solution with complexity $O(2^\lambda)$ to find a suitable $\overline{\text{pk}}$ to generate the desired $\tilde{E}_{\text{ch}}^{(1)}$ from sig_1 . The probability for this $\overline{\text{pk}}$ to generate the suitable $\tilde{E}_{\text{ch}}^{(2)}$ from sig_2 then has probability $2^{-\lambda}$, suggesting the overall complexity of $O(2^{2\lambda})$, which is infeasible. This motivates to formulate the following problem:

Problem 18 *Let E_1, E_2 be supersingular elliptic curves over $\mathbb{F}_{p^2}$. Let further $\text{sig}_1, \text{sig}_2$ be a pair of SQISIGN signatures generated for pk . Given E_1, E_2, sig_1 , and sig_2 , find a public key $\overline{\text{pk}}$ such that $\text{Cmt.R}(\overline{\text{pk}}, \text{sig}_1)$ recovers the challenge curve $\tilde{E}_{\text{ch}}^{(1)} = E_1$, and $\text{Cmt.R}(\overline{\text{pk}}, \text{sig}_2)$ recovers the challenge curve $\tilde{E}_{\text{ch}}^{(2)} = E_2$.*

Assuming Problem 18 to be hard, the probability for $\mathcal{A}$ to find a commitment collision for two given signatures is at most $\frac{1}{p} \approx \frac{1}{2^{2\lambda}}$, following the discussion above. For q_S given signatures, this amounts to the following heuristic result. We summarize the concrete S-UEO bounds for SQISIGN in Table 3.

Heuristic 19 *Let SQISIGN be the signature scheme as described above and assume Problem 18 to be hard. Then for any adversary $\mathcal{A}$, making q_S signing queries and q_H random oracle queries, it holds that*

$$\mathbf{Adv}_{\text{SQISIGN}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{2} \frac{1}{2^{2\lambda}}.$$

Together with Theorem 10, it then follows that

$$\mathbf{Adv}_{\text{SQISIGN}}^{\text{S-UEO}}(\mathcal{A}) \leq \binom{q_S}{2} \frac{1}{2^{2\lambda}} + \frac{q_H}{2^{|\text{ch}|}}.$$

4.4 The Code-Based Signature Scheme LESS

Description. LESS [BBB⁺24b] is a code-based signature scheme and a round-2 candidate in NIST's additional call for post-quantum signature schemes. It

uses the Fiat-Shamir transform and, for our analysis, we consider LESS with key prefixing (note that the LESS specification does not apply key prefixing). However, in contrast to the schemes analyzed above, LESS does not strictly apply Fiat-Shamir in the challenge variant, but a modified version of it. This variant corresponds to what we will call *salted Fiat-Shamir (challenge variant)* $\$-FS_{ch}$ and is displayed in Fig. 9. It works similar to FS_{ch} but involves a randomly generated salt. This salt is additionally used in the generation of the commitment and also hashed alongside public key (remember that we consider key prefixing), commitment, and message. The signature then comprises the challenge and response—as is usual for the challenge variant—but also the salt.

Analysis. Recall that the challenge of proving S-UEO security for FS_{ch} was the fact that the adversary might find public keys that recover the same commitment for different signatures, thereby increasing its probability of winning per random oracle query. For the transcript variant FS_{tr} , this problem is nullified by the fact that the randomly generated commitments are part of the signature and hashed when recomputing the correct challenge. It turns out that the salt in $\$-FS_{ch}$ achieves a similar effect as the commitment did for FS_{tr} . This allows us to apply essentially the same proof, i.e., the proof of Theorem 8, to obtain a similar bound. This is done in the theorem below.

Theorem 20. *Let Σ be an identification protocol and H be a random oracle. Consider the signature scheme $\$-FS_{ch}[\Sigma, H]$ in Fig. 9. Then for any S-UEO adversary $\mathcal{A}$, making q_S queries to the signing oracle and q_H queries to H , we have*

$$\text{Adv}_{\$-FS_{ch}[\Sigma, H]}^{\text{S-UEO}}(\mathcal{A}) \leq \frac{q_S^2}{2|\mathcal{S}_{salt}|} + \frac{q_H}{2^{|\text{ch}|}}.$$

Proof. The proof works essentially like the one for FS_{tr} (cf. Theorem 8). The difference is that we separate the winning probability of the adversary not based on a collision of the commitments but a collision of the salts. This changes the first term of the bound to a collision bound over the size of the salt space rather than the commitment space. The second term remains unchanged. $\square$

Applying this theorem to LESS directly yields the following bound for its S-UEO security.²⁰

Corollary 21. *Let LESS be the signature scheme as described in [BBB⁺24a]. Then for any adversary $\mathcal{A}$, making q_S signing queries and q_H queries to H (random oracle from the Fiat-Shamir transform), it holds that*

$$\text{Adv}_{\text{LESS}}^{\text{S-UEO}}(\mathcal{A}) \leq \frac{q_S^2}{2^{2\lambda+1}} + \frac{q_H}{2^\lambda}.$$

Table 4 shows S-UEO bounds for the parameter sets of LESS. As can be seen from the values, we obtain essentially λ -bit S-UEO security.

²⁰Note that the BUFF security of LESS was also analyzed in [ADM⁺24], however, this does not cover the variant of LESS with key prefixing that we consider.

$\$-FS_{ch}.Sign(sk, msg)$	$\$-FS_{ch}.Verify(pk, msg, sig)$
$salt \leftarrow \$ \mathcal{S}_{salt}$	$(salt, ch, rsp) \leftarrow sig$
$cmt \leftarrow \$ Cmt.G(sk, salt)$	$cmt \leftarrow Cmt.R(pk, salt, ch, rsp)$
$ch \leftarrow H(pk, cmt, salt, msg)$	if $cmt = \perp$
$rsp \leftarrow Rsp.G(sk, cmt, salt, ch)$	return 0
$sig \leftarrow (salt, ch, rsp)$	if $ch \neq H(pk, cmt, salt, msg)$
return sig	return 0
	return 1

Fig. 9: The salted Fiat-Shamir (challenge variant) transform. The difference is the usage of a randomly generated salt. Note that for $\mathcal{S}_{salt} = \emptyset$ the transform degenerates to FS_{ch} .

Table 4: Bounds for the S-UEO advantage of the LESS parameter sets.

	λ	$ \mathcal{S}_{salt} $	$\mathbf{Adv}_{LESS}^{S-UEO}(\mathcal{A})$
LESS-252- $\{192, 68, 45\}$ (NIST Level I)	128	2^{256}	$\leq \frac{q_S^2}{2^{257}} + \frac{q_H}{2^{128}}$
LESS-400- $\{220, 102\}$ (NIST Level III)	192	2^{384}	$\leq \frac{q_S^2}{2^{385}} + \frac{q_H}{2^{192}}$
LESS-548- $\{345, 137\}$ (NIST Level V)	256	2^{512}	$\leq \frac{q_S^2}{2^{513}} + \frac{q_H}{2^{256}}$

5 Conclusion

We showed that Fiat-Shamir signatures can achieve significantly better S-UEO bounds than what was shown in prior works. In particular, using 128-bit hash values for the challenges does not limit S-UEO security to 64-bits, which can be considered too little for practical purposes. This is especially relevant for SQUISH which, until now, was only shown to achieve 64-bit of S-UEO security due to a 128-bit hash output. We proved that our results are widely applicable by applying them to several signature schemes: Schnorr signatures, ML-DSA [NIST24a], SQUISH [CSD⁺23], LESS [BBB⁺24b], and CSI-FISH [BKV19]. Our work also leaves a few open problems for future work as described below.

While the practical relevance of M-S-UEO is debatable, our counterexample to rule out better bounds for M-S-UEO security is very contrived due to the prerequisite of weak public keys which accept everything. An open question is whether one can still achieve better M-S-UEO security by using additional assumptions which, at least, rule out weak keys. A similar, and more practically relevant, question is whether the same also holds for the S-UEO security of FS_{ct} for which our counterexample also relies on weak keys.

Regarding SQISIGN, our heuristic results rely on some additional, plausible assumptions. A formal analysis of these assumptions is another open question from our work.

Finally, for CSI-FISH, we have seen that our results do not apply to all parameter sets. We relied on the fact that there are more rounds than public key curves. We believe that our results also extend to other parameter sets where this is not the case, though one needs a more fine-grained analysis in this case.

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A Additional Analysis of Real-World Signature Schemes

We analyze the S-UEO security of CSI-FISH in [Appendix A.1](#).

A.1 The Isogeny-Based Signature Scheme CSI-FISH

CSI-FISH is a signature scheme based on isogenies between supersingular elliptic curves [\[BKV19\]](#). It applies the Fiat-Shamir transform in the challenge variant to an isogeny-based identification scheme. We briefly describe the relevant background, and refer to [\[BKV19, CLM⁺18\]](#) for more details. Note that in contrast to [\[BKV19\]](#), we consider a straightforward variant of CSI-FISH using key prefixing.

CSI-FISH is based on the commutative CSIDH class group action [\[CLM⁺18\]](#). It uses a set of supersingular elliptic curves $\mathcal{E}$ over $\mathbb{F}_p$ with a prime p , whose $\mathbb{F}_p$ -rational endomorphism ring is isomorphic to an order $\mathcal{O}(\sqrt{-p})$. The ideal class group $\text{cl}(\mathcal{O})$ of $\mathcal{O}$ is defined as the quotient of the fractional and principal ideals of $\mathcal{O}$. It acts freely and transitively on $\mathcal{E}$, i.e., there is a class group action $\text{cl}(\mathcal{O}) \times \mathcal{E} \rightarrow \mathcal{E}$. For any $\mathfrak{a} \in \text{cl}(\mathcal{O})$ and $E \in \mathcal{E}$, we write this action as $\mathfrak{a} * E = E'$, with $E' \in \mathcal{E}$. CSIDH describes a way to efficiently sample elements $\mathfrak{a} \in \text{cl}(\mathcal{O})$ and compute its action via isogenies (see [\[CLM⁺18\]](#)).

To set up CSI-FISH, [\[BKV19\]](#) computes the cardinality and structure of the ideal class group and sets $N = \#\text{cl}(\mathcal{O}) = \#\mathcal{E} \approx \sqrt{p}$. Given a generator $\mathfrak{l}$ of $\text{cl}(\mathcal{O})$, sampling elements then reduces to sampling an integer $a \in \mathbb{Z}/N\mathbb{Z}$ and setting $\mathfrak{a} = \mathfrak{l}^a$. CSI-FISH further provides a way of transforming $\mathfrak{a}$ into an equivalent ideal whose action can be evaluated efficiently. Thus, we fix $\mathfrak{l}$ and write $[a]E$ for the action $\mathfrak{l}^a * E$. Note that computing the structure of $\text{cl}(\mathcal{O})$ is infeasible for parameters exceeding CSIDH-512, where $\log_2 p \approx 512$ and $N \approx 2^{256}$, targeting $1^\lambda = 128$. However, the quantum security of this parameter set is debated [\[BLMP19, Pei20\]](#). Using SCALLOP [\[DFK⁺23\]](#) allows for increasing the size of p , but has a severe impact on the performance.

It is important to note that any element $E \in \mathcal{E}$ has a unique representation, and there is the following one-to-one correspondence: Given E and $a \in \mathbb{Z}/N\mathbb{Z}$, $E' = [a]E$ is the unique output of the group action. Conversely, given $E, E' \in \mathcal{E}$, there is a unique $a \in \mathbb{Z}/N\mathbb{Z}$ satisfying $E' = [a]E$.

CSI-FISH with one-bit challenges. CSI-FISH defines a signature scheme based on an identification scheme using this action. The basic version of this identification scheme, using 1-bit challenges, is depicted in [Fig. 10](#). We fix a starting curve E_0 , and the prover samples a secret key $a \in \mathbb{Z}/N\mathbb{Z}$ and computes the public key $E_1 = [a]E_0$. They commit to a curve $E^{(b)}$ through sampling a random $b \in \mathbb{Z}/N\mathbb{Z}$ and computing $E^{(b)} = [b]E_0$. The verifier samples a random challenge $c \in \{0, 1\}$ and sends it to the prover, who responds with $r = b - ca$. The verifier then checks if $[r]E_c = E^{(b)}$. By repeating this procedure λ times, for security parameter λ , and applying the Fiat-Shamir transform (challenge variant), we obtain the CSI-FISH signature scheme, as shown in [Fig. 11](#).

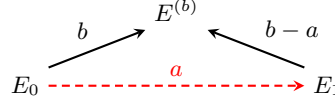


Fig. 10: Illustration of ID scheme underlying CSI-FISH. E_0 is the public starting curve, E_1 is the public key curve, and a is the secret key. The commitment curve is $E^{(b)}$ generated by choosing b at random. Depending on the challenge c , the response is $r = b$ (if $c = 0$) or $r = b - a$ (if $c = 1$).

Theorem 22. *Let CSI-FISH be the signature scheme as shown in Fig. 11. Then for any adversary $\mathcal{A}$, making q_S signing queries and q_H random oracle queries, it holds that*

$$\text{Adv}_{\text{CSI-FISH}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{2} \frac{1}{N^{\lambda-1}}.$$

Proof. Let $\text{sig}_k = (r_1^{(k)}, \dots, r_\lambda^{(k)}, c_1^{(k)}, \dots, c_\lambda^{(k)})$ for $1 \leq k \leq q_S$ be the q_S signatures that $\mathcal{A}$ receives from its oracle, for messages msg_k of its choice. Since the individual responses are computed as $r = b - ca$, for uniform random b , the responses are distributed uniformly over $\mathbb{Z}/N\mathbb{Z}$. For any distinct pair $\text{sig}_i, \text{sig}_j$, to obtain the same commitment for a curve E_1 of $\mathcal{A}$'s choice, the following must hold for all $1 \leq k \leq \lambda$:

$$[r_k^{(i)}]E_{c_k^{(i)}} = [r_k^{(j)}]E_{c_k^{(j)}} = E^{(k)}. \quad (4)$$

We distinguish between two cases:

Case 1: $\forall k : c_k^{(i)} = c_k^{(j)}$ (challenge bits of sig_i and sig_j are equal).

In this case, Eq. (4) holds if $r_k^{(i)} = r_k^{(j)}$ for all k . Due to the uniform distribution of the responses, this happens with probability $(1/N)^\lambda$, independent of the chosen E_1 .

Case 2: $\exists k : c_k^{(i)} \neq c_k^{(j)}$ (challenge bits of sig_i and sig_j are different).

In this case, let $k' \in [\lambda]$ be such that $c_{k'}^{(i)} \neq c_{k'}^{(j)}$, which is guaranteed to exist by assumption. Wlog assume that $c_{k'}^{(i)} = 0$ (swap the signatures otherwise). In order to satisfy Eq. (4), $\mathcal{A}$ has to choose E_1 by computing $E^{(k')} = [r_{k'}^{(i)}]E_0$, and setting $E_1 = [-r_{k'}^{(j)}]E^{(k')}$. For all $k \neq k'$, we distinguish two additional cases:

Case 2.1: the k -th challenge bits agree, i.e., $c_k^{(i)} = c_k^{(j)}$.

If $c_k^{(i)} = c_k^{(j)}$, then Eq. (4) can only be satisfied if $r_k^{(i)} = r_k^{(j)}$, which has probability $1/N$.

Case 2.2: the k -th challenge bits disagree, i.e., $c_k^{(i)} \neq c_k^{(j)}$.

If $c_k^{(i)} \neq c_k^{(j)}$, since E_0 and E_1 are already fixed, we require suitable $r_k^{(i)}, r_k^{(j)}$ such that Eq. (4) is satisfied, again with a probability of $1/N$.

KeyGen()	Sign(sk, msg)
$a \leftarrow \$_{\mathbb{Z}/N\mathbb{Z}}$	$a \leftarrow \text{sk}$
$E_1 \leftarrow [a]E_0$	$b_1, \dots, b_\lambda \leftarrow \$_{\mathbb{Z}/N\mathbb{Z}}$
$\text{sk} \leftarrow a$	for $i = 1.. \lambda$
$\text{pk} \leftarrow E_1$	$E^{(i)} \leftarrow [b_i]E_0$
return (sk, pk)	$(c_1, \dots, c_\lambda) \leftarrow \text{H}(E_1, E^{(1)}, \dots, E^{(\lambda)}, \text{msg})$
	for $i = 1.. \lambda$
	$r_i \leftarrow b_i - c_i a$
	$\text{sig} \leftarrow (r_1, \dots, r_\lambda, c_1, \dots, c_\lambda)$
	return sig
Cmt.R(pk, sig)	Verify(pk, msg, sig)
$E_1 \leftarrow \text{pk}$	$(r_1, \dots, r_\lambda, c_1, \dots, c_\lambda) \leftarrow \text{sig}$
$(r_1, \dots, r_\lambda, c_1, \dots, c_\lambda) \leftarrow \text{sig}$	$(E^{(1)}, \dots, E^{(\lambda)}) \leftarrow \text{Cmt.R}(\text{pk}, \text{sig})$
for $i = 1.. \lambda$	if $\text{H}(E_1, E^{(1)}, \dots, E^{(\lambda)}, \text{msg}) \neq (c_1, \dots, c_\lambda)$
$E^{(i)} \leftarrow [r_i]E_{c_i}$	return 0
return $(E^{(1)}, \dots, E^{(\lambda)})$	return 1

Fig. 11: Fiat-Shamir signature scheme CSI-FISH (including key prefixing) with an explicit description of the commitment recovery algorithm Cmt.R.

Since there are $\lambda - 1$ indices $k \neq k'$, this leads to a total probability of a 2-commitment collision of $(1/N)^{\lambda-1}$.

From the above, we conclude that $\mathcal{A}$ can find a public key leading to a 2-commitment collision for two arbitrary, fixed signatures with probability at most $(\frac{1}{N})^{\lambda-1}$. Overall, since the adversary obtains q_S signatures, it holds that

$$\text{Adv}_{\text{CSI-FISH}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{2} \frac{1}{N^{\lambda-1}},$$

which finishes the proof. $\square$

The corollary below provides a bound on the S-UEO advantage of an adversary against the CSI-FISH signature scheme. It follows directly from [Theorem 10](#) and [Theorem 22](#). We provide specific values for bound in [Table 5](#).

Corollary 23. *Let CSI-FISH be the signature scheme as shown in [Fig. 11](#). Then for any adversary $\mathcal{A}$, making q_S and q_H signing queries and random oracle queries, respectively, it holds that*

$$\text{Adv}_{\text{CSI-FISH}}^{\text{S-UEO}}(\mathcal{A}) \leq \binom{q_S}{2} \frac{1}{N^{\lambda-1}} + \frac{q_H}{2^{\text{ch}}}.$$

Table 5: S-UEO advantage for CSI-FISH using the CSIDH-512 parameter set. As can be seen, CSI-FISH effectively achieves the ideal bound $\frac{q_H}{2^{\lceil \text{ch} \rceil}}$.

λ	t	q_S	$\text{Adv}_{\text{CSI-FISH}}^{\text{S-UEO}}(\mathcal{A})$
128	2	2^{48}	$\leq \frac{1}{2^{32417}} + \frac{q_H}{2^{\lceil \text{ch} \rceil}}$
		2^{64}	$\leq \frac{1}{2^{32385}} + \frac{q_H}{2^{\lceil \text{ch} \rceil}}$

CSI-FISH with larger challenge space. Instead of a single public key curve E_1 and performing $R = \lambda$ rounds with 1-bit challenges $c_i \in \{0, 1\}$, an increased public key size and challenge space can reduce the number of rounds. We refer to the modified scheme as CSI-FISH_{LCS}. We fix $S > 1$, sample a secret key as $a_1, \dots, a_{S-1} \in \mathbb{Z}/N\mathbb{Z}$ and compute public key curves $E_i = [a_i]E_0$ for $1 \leq i \leq S-1$. This allows for using challenges $c_i \in \{0, \dots, S-1\}$, when computing responses as $r_i = b_i - s_{c_i}$, where we define $s_0 = 0$. With the reduced soundness error of $1/(S-1)$, this allows for decreasing the number of rounds R of the identification protocol.²¹

Intuitively, more public key curves and fewer rounds give an adversary more freedom to find t -commitment collisions. Indeed, this highly depends on the choice of S and R . In the following, we focus on the CSI-FISH parameters that optimize for small combined public key and signature sizes. For these parameter sets, we have $S < R$ (cf. [BKV19, Table 3]). The following theorem shows that for these cases, the probability for finding t -commitment collisions is negligible.

Theorem 24. *Let CSI-FISH_{LCS} be the signature scheme described above, and let $S < R$. Then for any adversary $\mathcal{A}$, making q_S signing queries and q_H random oracle queries, it holds that*

$$\text{Adv}_{\text{CSI-FISH}_{\text{LCS}}}^{2\text{-Cmt-CR}_r}(\mathcal{A}) \leq \binom{q_S}{2} \frac{1}{N^{R-S}}.$$

Proof. Let $\text{sig}_k = (r_1^{(k)}, \dots, r_\lambda^{(k)}, c_1^{(k)}, \dots, c_\lambda^{(k)})$ for $1 \leq k \leq q_S$ be the q_S signatures that $\mathcal{A}$ receives from its oracle, for messages msg_k of its choice. Let sig_i and sig_j be any distinct pair of these signatures. $\mathcal{A}$ has to find a public key $E_1, \dots, E_{S-1}$ ²² such that

$$[r_k^{(i)}]E_{c_k^{(i)}} = [r_k^{(j)}]E_{c_k^{(j)}} = E^{(k)} \quad (5)$$

²¹CSI-FISH_{LCS} further uses larger challenge spaces through additionally using twists of public key curves, and slow hashing to further reduce R . Beyond their influence on the choice of S and R , these optimizations are irrelevant for our analysis, and we omit their description for the sake of simplicity.

²²Note that game 2-Cmt-CR_r requires the adversary to find a key *different* from the given public key. Our bound holds even if that requirement is dropped which is why we do not check for different public keys.

for all $1 \leq k \leq R$, in order to obtain the same commitment. We will show that any assignment of public key curves leads to a 2-commitment collision with probability of at most $(1/N)^{R-S}$ by first presenting an assignment strategy and then generalizing it.

We start by picking a random index $1 \leq k \leq R$ and assigning $E_{c_k^{(i)}}$ and $E_{c_k^{(j)}}$, by arbitrarily choosing one curve and fixing the second curve to satisfy Eq. (5).²³ If $c_k^{(i)} = 0$ or $c_k^{(j)} = 0$, we use the fixed E_0 and assign the other curve as above, such that Eq. (5) is satisfied. In case $c_k^{(i)} = c_k^{(j)}$, we obtain a success probability of $1/N$ for Eq. (5) to hold, independent of the chosen curve.

In the next step, we look for all other occurrences of $c_k^{(i)}$ or $c_k^{(j)}$ at other indices $k' \neq k$, e.g., where $c_{k'}^{(i)} = c_k^{(i)}$ or $c_{k'}^{(j)} = c_k^{(j)}$. Wlog, let $c_{k'}^{(i)} = c_k^{(i)}$, if such an index exists. Since $E_{c_k^{(i)}} = E_{c_{k'}^{(i)}}$ is fixed, we can either fix $E_{c_{k'}^{(j)}}$ such that Eq. (5) is satisfied, or if $E_{c_{k'}^{(j)}}$ is already fixed, obtain a probability of $1/N$ for a commitment collision at index k' .

We iteratively repeat this approach, until we covered all indices k for which any of the challenges $c_k^{(i)}$ and $c_k^{(j)}$ already occurred, and the corresponding public key curves have been fixed. Since $S < R$, we either have at least one index with collision probability of $1/N$, or we have not covered all R indices. In the latter case, we restart the same approach from one of the remaining indices. Repeating this until all indices have been covered, we will obtain at least $R - S$ indices with a commitment collision probability of $1/N$.

Note that the arbitrary choices of curves E_c at the beginning of each iterative process do not have an influence on whether a collision occurs or not. In particular, shifting the arbitrary choice of a curve E_c by the action of some $[a']$ shifts all other curves in the following assignments by $[a']$, too. Hence, either a collision with probability $1/N$ occurs for every arbitrary choice, or for none. Furthermore, multiple arbitrary assignments of curves have no influence on one another. Similarly, the same curves are fixed during an iteration, independent of the index k that was used for the initial assignment. Therefore, any assignment of curves will lead to the same 2-commitment-collision probability of at most $(1/N)^{R-S}$ for any given pair of signatures $\mathbf{sig}_i$ and $\mathbf{sig}_j$.

Since $\mathcal{A}$ receives q_S signatures, this probability multiplied by $\binom{q_S}{2}$ gives the upper bound and concludes the proof. $\square$

²³We will explain below why this arbitrary choice of curves and indices has no implications on whether a collision occurs.